VISITOR'S LOG MONITORING SYSTEM AND CONTRABAND DETECTION IN CORRECTIONAL INSTITUTION

AN UNDERGRADUATE THESIS

Presented to the Faculty of
College of Computing Studies
ZAMBOANGA DEL SUR PROVINCIAL GOVERNMENT COLLEGE
MAIN CAMPUS

In partial fulfillment
of the requirements for the degree
BACHELOR OF SCIENCE IN INFORMATION SYSTEMS
(BSIS)

JANDEL A. VILLA
DULCE MARIE C. MAGLASANG

December 2024

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CERTIFICATE OF PANEL APPROVAL

The thesis entitled **"VISITOR'S LOG MONITORING SYSTEM AND CONTRABAND DETECTION IN CORRECTIONAL INSTITUTION"**, prepared and submitted by JANDEL A. VILLA and DULCE MARIE V. MAGLASANG in partial fulfillment of the requirements for the degree Bachelor of Science in Information Systems, is hereby recommended for approval.

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DEDICATION

This thesis is respectfully dedicated to our Almighty God, whose divine guidance and blessings have been our constant source of strength and wisdom. To our parents and families, whose unwavering support, love, and sacrifices have made this journey possible. To our adviser and mentors, for their invaluable guidance and expertise that greatly contributed to the completion of this work. And to all individuals who have inspired and supported us in any way, we humbly dedicate this work as a token of our deepest gratitude and respect.

ABSTRACT

VILLA, JANDEL A., MAGLASANG, DULCE MARIE V. College of Computing Studies, Zamboanga del Sur Provincial Government College, Aurora, Zamboanga del Sur, December 2024. **VISITOR'S LOG MONITORING SYSTEM AND CONTRABAND DETECTION IN CORRECTIONAL INSTITUTION.** An unpublished Bachelor's Thesis.
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This study presents the development of a Visitor's Log Monitoring System with Contraband Detection designed for correctional institutions. The system addresses the need for efficient visitor management and enhanced security by leveraging image processing technology for contraband detection. The primary objectives of the system are to monitor and manage visitor logs, streamline inmate visitation, and detect prohibited items in compliance with the institution's manual of operations. Using a camera integrated with image processing algorithms, the system identifies and alerts staff to contraband items brought by visitors or staff during visitation. The contraband detection process is automated and trained to recognize predefined prohibited items based on a database provided by the institution. The methodology emphasizes a seamless user experience with a user-friendly interface for managing visitor records. By integrating an image processing module for contraband scanning, the system enhances security with faster, more accurate detection than manual methods. Additionally, alert mechanisms ensure timely notifications of security threats, enabling quick staff response. Testing results show improved accuracy and efficiency, significantly reducing human error typical of manual checks. These advancements strengthen institutional security and ensure a more reliable environment. This research contributes to modernizing correctional facility operations by integrating advanced technologies, ensuring operational efficiency and adherence to security protocols. Future improvements may include expanding the contraband database and incorporating machine learning for adaptive detection.

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CHAPTER I

THE PROBLEM

Introduction

These days the development of modern world is fast due of having advancement in technology which rapidly seeking towards success. One of the useful modern technologies that have been created is an "Automated or Computerize System". It is now used by various fields such as institutions, business, communication, companies, science and even the learning of process of old-fashioned way was shifted to technology ways. System exists to these fields to help access information and processes easier and more productive.

This study aims to design and develop a Visitors Log Monitoring System and Contraband Detection specifically for correctional institutions. The purpose of this system is to enable accurate and reliable monitoring of visitors as well as the detection of prohibited items entering correctional facilities. The system will provide authorized personnel with efficient data management capabilities, including secure backup of information stored on computers located in the guardhouse. Key features will include tracking of visitor log-in and log-out times, management of visitor information, administrative access controls, a database backup module, and detailed records of inmates, visitors, and contraband items.

Prison security has become an increasingly critical concern at local, national, and international levels. With rising incarceration rates, correctional institutions face ongoing challenges related to the smuggling of contraband, which poses significant risks to the safety of both staff and inmates. Despite various preventative measures, determined individuals continue to find innovative ways to circumvent security protocols, particularly during inmate visitation periods. Research by Gangi (2017) highlights that correctional facilities remain vulnerable to contraband infiltration during visitations, even with strengthened staff efforts to prevent such incidents.

Contraband items including drugs, weapons, and illicit communication devices threaten institutional security by enabling violence, addiction, and organized criminal activity within prison environments. Traditional security measures such as pat-downs and strip searches are widely used but often have limited effectiveness (Shukla et al., 2021). Although whole body scanners offer a more reliable method for detecting concealed contraband, their use is constrained by high costs, health regulations, and operational challenges. Correctional officers operate under considerable pressure to enforce anti-contraband policies while ensuring a safe environment for all individuals within the facility. However, limited research exists regarding officers' perceptions of safety and the difficulties they encounter in enforcing these protocols. Tititampruk and Ketsil (2021) emphasize that understanding officer perspectives is vital for developing more effective safety strategies in correctional settings.

Visitor management is a fundamental component of correctional security. While visitation provides an important connection between inmates and the outside world, it is frequently exploited as a means for contraband smuggling. Many correctional institutions continue to rely on traditional paper logbooks, which are susceptible to errors, inefficiencies, and security breaches. Oktaviandri and Foong (2019) argue that implementing a digital Visitor Management System enhances the accuracy of records, improves administrative efficiency, and reduces security vulnerabilities, especially in facilities with high visitor volume.

The effectiveness of modern contraband detection technologies varies depending on the prison's physical layout, capacity, and operational policies. Dix et al. (2021) state that the successful deployment of such technologies requires consideration of operational feasibility, legal compliance, and health safety. Technologies must be adaptable to detect contraband concealed in different ways, such as within food, mail, clothing, or on the body. This highlights the necessity for a flexible, intelligent, and scalable monitoring system.

This study highlights the pressing need to enhance security protocols within correctional facilities through the implementation of a comprehensive and technology-driven solution. Correctional institutions continue to face significant challenges in monitoring visitor activity and preventing the introduction of unauthorized items that may compromise safety. The research addresses these concerns by designing and implementing an integrated system capable of accurately recording visitor information while simultaneously incorporating mechanisms for detecting potential contraband during visitations.

The primary goal is to strengthen safety protocols and provide a more reliable method of monitoring interactions between inmates and visitors. By doing so, the system helps reduce opportunities for security breaches and supports the overall order and stability of the facility. In addition, this study emphasizes how the application of modern technology can significantly enhance the efficiency and responsiveness of correctional staff, ultimately ensuring a safer environment for inmates, personnel, and all other stakeholders. Through this innovative solution, the project seeks to contribute to lasting improvements in correctional security and operational standards.

Objectives of the study

This study addresses critical challenges related to visitor monitoring and contraband detection within correctional facilities. By utilizing technological advancements, specifically image processing and offline web-based systems, the research aims to develop a secure and efficient solution tailored to the operational needs of these institutions. The objectives outlined below serve as the foundation for the development and implementation of the proposed system.

General Objective

To develop a comprehensive Visitor's Log Monitoring System and Contraband Detection platform that utilizes image processing technologies and operates as an offline website to enhance security and streamline visitor management in correctional institutions.

Specific Objectives

These specific objectives are based on the issues identified in the introduction regarding visitor monitoring and contraband detection, aiming to enhance security and operational efficiency in correctional facilities through technological solutions. It aims to achieve the following:

1. To develop a streamlined visitor log monitoring system that accurately records visitor entries and exits, ensuring efficient tracking and identification of potential security risks.
2. To implement an image-processing-based contraband detection system using Google Cloud Vision to enhance the accuracy and effectiveness of identifying prohibited items.
3. To improve visitor management and security protocols by integrating an automated system that enhances service efficiency and reduces security threats within the correctional facility.

Significance of the Study

This study contributes significantly to the advancement of security measures in correctional institutions, particularly in the management of visitor logs and contraband detection. By integrating image processing technology, the system provides a cost-effective and easily deployable solution to improve the detection of prohibited items during visitations. This study will be beneficial to the following;

Correctional Institutions. The system enhances security by providing accurate visitor tracking and real-time contraband detection, reducing unauthorized access and prohibited item entry.

Correctional Staff and Administrators. Staff members benefit from a more efficient visitor management process, reducing manual workload and improving security enforcement. Administrators gain a reliable system for monitoring visitor activities and managing security violations effectively.

Inmates. Ensuring a secure environment protects inmates from potential threats posed by contraband smuggling and unauthorized visitors.

Visitors. A well-structured logging system enhances visitor experience by minimizing delays and ensuring fair and secure visitation procedures.

Law Enforcement and Government Agencies. The system provides reliable documentation of visitor activities and contraband incidents, assisting in policy enforcement and investigations.

Future Researcher. The researchers gained additional knowledge and insights from the proposed study, which facilitated their ability to find references, particularly when conducting related research. This made it more convenient for them to seek sources for their studies.

Scope and Limitation

This study focuses on developing a Visitor's Log Monitoring System and Contraband Detection in Ramon Magsaysay District Jail, Ramon Magsaysay Zamboanga del Sur to enhance security and efficiency in correctional facilities. The system allows administrator and staff to track visitor entries and exits, ensuring accurate records while also using image processing technology to scan and detect prohibited items through a mobile camera. It will store visitor data, generate reports, and provide role-based access for authorized personnel. However, the system has some limitations. Since it operates offline, it won’t support real-time cloud updates or remote access. Its contraband detection accuracy depends on image quality, lighting, and AI recognition capabilities, meaning manual verification by staff is still necessary. Additionally, the system does not include biometric verification like fingerprint or facial recognition and requires a mobile device with a camera for scanning. Despite these limitations, the system aims to improve visitor monitoring, enhance contraband detection, and create a safer environment for inmates, staff, and visitors.

Definition of Terms

The following section defines key terms used throughout the study; these definitions aim to help readers and future researchers better understand the technical concepts and context of the system discussed:

Alert Notification. An automated system message or signal that notifies security personnel of detected threats, such as the presence of contraband.

Authentication. A verification process that confirms the identity of users before granting access to the system, typically through login credentials such as usernames and passwords.

Conjugal Visit. A private visit granted to an inmate, allowing them to spend time alone with their legal spouse or partner in a controlled environment, usually within the prison grounds.

Contraband Detection. A security feature that uses image processing technology to identify and flag prohibited items that visitors or personnel may attempt to bring into the facility.

Correctional Institution. A facility such as a jail or prison where individuals are confined as part of their legal sentence or while awaiting trial.

Data Encryption. A cybersecurity measure that converts sensitive data into a secure, unreadable format to prevent unauthorized access or data breaches.

Google Cloud Vision. A cloud-based AI tool developed by Google that analyzes images using machine learning. It can detect, classify, and extract information such as objects, text, or patterns. In this system, it scans visitor belongings and identifies potential contraband by comparing images with a predefined database of prohibited items.

Image Processing. A technique used in computer vision where images are analyzed and interpreted by software to extract meaningful information, such as identifying contraband items.

Inmate. A person who is confined or held in a correctional facility, such as a jail, prison, or detention center, usually as a result of being accused or convicted of a crime.

Inmate Monitoring. The process of observing inmate activities and interactions to ensure compliance with institutional rules and to maintain overall safety and security.

Legal Visit. A private meeting between an inmate and their lawyer or legal representative.

Offline System. A system that operates without internet connectivity, meaning all data is stored and processed locally within the correctional facility's network.

Prohibited Items (Contraband). Any unauthorized objects, such as weapons, drugs, or communication devices, that pose security risks within a correctional institution.

Person Deprived of Liberty (PDL). Refers to an individual who is confined in a detention or correctional facility as a result of legal proceedings or a court sentence. This term includes individuals awaiting trial, those serving sentences, or those under custodial investigation, and is used to promote dignity and human rights in the treatment of incarcerated individuals.

Real-Time Monitoring. The ability of the system to instantly track and display visitor movements and security alerts within the correctional facility.

Regular Visit. A standard visit from family members, relatives, or approved friends.

Religious Visit. A visit from a religious leader, chaplain, or spiritual adviser.

Role-Based Access Control (RBAC). A security mechanism that restricts system access based on user roles, ensuring that only authorized personnel can manage visitor logs and security alerts.

System Dashboard. A centralized visual display within the software that provides real-time information on system activities, visitor logs, and detection alerts.

Tele-Visit. A remote video visit using technology like video conferencing.

User Interface (UI). The part of the system that users interact with directly, designed to be intuitive and user-friendly for easier navigation and task completion.

Visitor Management System. A platform designed to streamline the process of registering, tracking, and managing visitors entering and leaving the facility.

Visitor Screening. A security procedure that involves checking visitors for contraband using scanning devices, image processing, and manual inspections.

Visitor Tracking. The process of recording and monitoring individuals who enter and exit the facility, including their identity, purpose of visit, and duration of stay.

Visitation Schedule. A predetermined timetable that outlines the specific days and times when inmates are allowed to receive visitors.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter reviews relevant local and international literature that highlights the ongoing challenges in visitor management and contraband detection within correctional facilities. It examines the limitations of traditional methods in addressing security concerns, particularly as smuggling tactics become more sophisticated. Through the analysis of previous studies, legal frameworks, and technological advancements, this section establishes the need for modern, efficient systems. These insights provide the basis for developing a technology-based solution to enhance safety and streamline operations in correctional institutions.

Local Study

One of the most pressing issues facing correctional facilities today is prison overcrowding, which significantly contributes to poor living conditions. It is widely regarded as the most critical challenge confronting prison systems, with consequences that can, in extreme cases, be life-threatening. Overcrowding often leads to increased incidents of violence, escape attempts, and the smuggling of contraband such as synthetic drugs, mobile phones, and even drones. The introduction of machine-based detection systems offers substantial support in maintaining peace and safety for both inmates and staff. These systems enhance monitoring capabilities and reduce the burden on correctional personnel.

Incorporating such technology is an important step toward modernizing security protocols and addressing long-standing operational deficiencies.

“*Steep penalties sought to end contraband smuggling in jails*

Philippine News Agency reporter Jose Cielito Reganit (2022) highlighted the confiscation of thousands of contraband items at the New Bilibid Prison (NBP). In response, Surigao del Norte Representative Robert Ace Barbers filed a bill aimed at ending the widespread entry of prohibited objects and items into correctional and detention facilities across the country. The persistent presence of contraband continues to be a long-standing issue within the national prison system, contributing to the deterioration of institutional order and safety. Barbers emphasized that the discovery of a middleman inmate who used a cellphone to hire gunmen in the killing of journalist Percy Lapid served as a key motivation to propose legislation addressing the ongoing problems in prison management, particularly concerning the infiltration of dangerous items and the coordination of criminal acts from within detention centers.

Diskarte Lang: Dealing with Operational Challenges in a Philippine City Jail.

According to Nario-Lopez, H. G. (2021), the subhuman conditions in Philippine jails and prisons remain a pressing concern due to the adverse impact on the delivery of rehabilitative justice. Although multiple factors contribute to the issue across various institutions involved in the justice process, focusing solely on prison administration and the role of jail officers may oversimplify the problem and place undue emphasis on the influence of prison regimes. Authority granted to officers by official position is often contested and negotiated by detainees, particularly in the context of persistent structural deficiencies within correctional facilities.

"Bucor relieves 7 for investigation over strip search"

According to a report by Philippine News Agency Benjamin Pulta (2024), seven officers were relieved of duties to allow for an investigation into a complaint filed by the wives of persons deprived of liberty (PDLs), who were recently subjected to a strip search at the New Bilibid Prison (NBP). A strip search refers to the procedure of inspecting an individual for weapons or contraband suspected to be concealed on the body or within clothing, often requiring the removal of some or all garments. This method continues to be enforced without exception across correctional institutions due to the increasing number of incidents involving visitors attempting to smuggle prohibited items concealed within body cavities. Bureau of Corrections Director General Gregorio Pio O. Catapang Jr. stated that, in the absence of body scanners, manual searches remain necessary. Efforts are ongoing to secure funding for the procurement of advanced screening equipment that would eliminate the need for physical contact during inspections.

Foreign Study

The ubiquity of correctional corruption stems, in part, from organizational factors that make correctional environments particularly susceptible to corrupt practices. It is important to recognize that correctional behavior does not occur within a vacuum: correctional corruption is not only a reflection of the choices of particular individuals, but must also be understood as behaviors that occur in the context of the organizational and normative systems within which individuals work.

Such a perspective is essential, especially where it is clear that instances of corruption are not isolated or that they recur over time. The correctional environment has its own unique features, in terms of organizational climates, structures, and cultures, which provide opportunities for deviance (Goldsmith, Halsey and Groves, 2016).

"Roundtable: 7 critical issues in corrections in 2018"

In an interview conducted by Nancy Perry (2018), Anthony Gangi emphasized that contraband remains one of the most serious threats within correctional environments. The discussion underscored the necessity of both proactive and reactive strategies to prevent the entry of prohibited items. Emphasis was placed on the importance of providing correctional institutions with proper detection tools, sufficient staffing, and a strong focus on officer safety to effectively manage contraband-related risks. Similarly, Joe Bouchard highlighted the role of the underground economy in triggering violence and injuries among staff within prison facilities. Recommendations included the implementation of basic contraband control protocols, adequate personnel deployment, and the adoption of advanced technological measures to mitigate the circulation of illicit materials.

Joe Russo, Dulani Woods, John S. Shaffer, and Brian A. Jackson (2019) stated that many of these threats risk public safety. In light of the ongoing challenges, the corrections sector faces in countering these threats, RAND researchers convened an expert workshop to better understand the challenges and identify the high-priority needs associated with threats to institutional security. Addressing the research need and developing the tools and resources as prioritized by the workshop participants is one route to providing correctional institutions the support needed to confront security threats going forward.

"House bill aiming to block entry of contrabands in jails advances"

According to Gabriel Pabico Lalu (2023), The House of Representatives has approved on third reading a bill that would mandate creating a system that deters entry of contraband items like drugs and gadgets into custodial and detention facilities across the country. The CDCS shall include modern technology, devices, or units, such as handheld and walk-through metal detectors, X-ray scanners, and K9 units.

Visitors and employees also smuggle wireless phones and related paraphernalia into prisons. In some cases, staff members have accepted bribes, usually for several hundred dollars per device, from inmates to sneak cell phones into facilities. For example, one correctional officer reported earning more than $100,000 by charging prisoners $100 to $400 per device. Smuggled wireless phones also provide a source of additional income to inmates who charge other prisoners up to $50 for each call placed (Burke, T. W., and Owen, S. S. 2010)

First, correctional institutions must maintain vigilance in detecting the smuggling and possession of devices by prisoners, utilizing both traditional and innovative methods, such as the use of canines. Second, should the legality of jamming technologies be established, research should focus on criminalizing inmate cell phone possession rather than treating it as standard contraband; a proposal for such legislation has been introduced in California. Finally, authorities should promote the use of technology as a preventive and decision-making tool for managing cell phone contraband within prisons and jails. The increasing number of smuggled devices highlights the urgency of addressing this issue (Ochola, 2015).

Visitors engaged in different strategies to retain the perceived loss of privacy. For many, this involved waiting in the car until visitation time, rather than in the waiting room. One strategy involved the importation of cell phones, laptop computers, and pens/paper as a crutch or support for the visit. These are all banned items, which produced resentment in visitors (Sitren, et. al 2021).

Russo, J., and Wells, D. (2019). As long as correctional institutions have existed, there has been contraband. Contraband can be a moving target in more ways than one. What is considered contraband may vary among correctional agencies and over time. Some types of contraband are consistent across jurisdictions (e.g., weapons, tools, drugs, alcohol). Some contraband may vary among jurisdictions (e.g., tobacco was sold in institutions for decades but is now considered contraband in the states that have banned smoking). Further, societal and technological changes may create new forms of contraband. For example, 25 years ago, cellphones were not thought of as a threat to correctional institutions as they are today. Based on case studies of one prison and two jail systems in the US, the typical interdiction process includes processing mail through x-ray machines, running every person (visitor, staff member, and resident) entering the facility through a metal detector or body scanner, regularly patrolling the perimeters of facilities, and conducting both targeted and random cell searches and shakedowns (Shukla, Peterson, and Kim 2021). Additionally, permissible items may be considered contraband if an individual is in possession of more than the allotted amount (e.g. towels might be considered contraband if an individual has three while the facility only permits the possession of two), (Shukla et al., 2021).

Google Cloud Vision API empowers experts to comprehend an object in an image or a photo by speaking to fruitful machine learning models in a simple to utilize REST API. It rapidly characterizes pictures into a large number of classifications (e.g., "sailboat", "Eiffel Tower"), distinguishes singular questions and faces inside pictures, and finds printed words contained inside pictures. One can build metadata on picture index, direct hostile content, or empower new advertising scenarios through picture assessment investigation. Analyze images uploaded in the request or integrate with your image storage on Google Cloud Storage (Rajwani, R., et al., 2018) and a successful implementation of contraband detection technology must consider three common challenges: operational achievability to consider how technology adoption may work for each individual facility; policies and legislative constraints, which attempt to balance security and privacy; and health and safety standards, which ensure individuals are not exposed to high levels of radiation. Key questions for each consideration are presented and can serve as a foundation for criminal justice leaders and decision makers as they evaluate potential impacts of detection technologies (CJTTEC, 2021).

Synthesis of the Study

The studies and literature review highlight key concerns faced by correctional facilities, particularly in managing contraband and maintaining security. Both local and foreign research emphasize the ongoing issue of contraband smuggling, the inadequacy of traditional detection methods, and the need for technological innovation to improve security.

Local studies link prison overcrowding to increased contraband smuggling, with items such as phones, synthetic drugs, and hacking tools fueling criminal activity. Legislative efforts are underway, but many institutions still rely on manual inspections due to limited access to modern scanning technology, leaving significant security gaps. Foreign research reflects similar concerns, identifying corruption, weak systems, and outdated tools as contributors to the problem. Experts advocate for adequate staffing, advanced scanning devices, and clear policies. Technologies like Google Cloud Vision and AI-powered detection offer promising solutions, but their implementation must be carefully managed, balancing legal and ethical considerations.

A common theme across all studies is the evolving nature of contraband threats. Items once considered harmless, such as cell phones, now pose significant risks. Addressing these challenges requires a multifaceted approach, combining advanced technology, effective policies, and necessary administrative reforms to stay ahead of emerging security threats.

In conclusion, the studies emphasize the urgent need to modernize contraband detection and prison management systems. By integrating AI tools and supporting regulatory frameworks, correctional facilities can improve safety, protect institutional integrity, and ensure the ethical use of emerging technologies.

CHAPTER III

TECHNICAL BACKGROUND

This chapter provides an elaborate discussion of the technical foundations pertinent to this study. It initiates with an assessment of the existing system, clarifying its strengths and limitations, and subsequently progresses to an extensive exploration of the proposed system, including its hardware and software prerequisites.

Overview of the Current System

The current system for managing visitors, inmates, and contraband detection in correctional institutions is based on the Manual of Operations, which outlines protocols for visitor registration, contraband inspection, and inmate management. While these procedures are designed to maintain security and order, they remain manual and time-consuming, leading to inefficiencies in enforcement and contraband control.

Visitor screening and contraband detection follow strict protocols, including physical searches, documentation of confiscated items, and adherence to ethical guidelines. A zero-tolerance policy ensures that search personnel conduct inspections professionally and properly record incidents. Visitors and personnel undergo body and strip searches in compliance with legal and human rights standards, while vehicle inspections help prevent the smuggling of prohibited items. Non-compliance results in denied entry for visitors or disciplinary actions for personnel.

Despite these measures, the lack of technological integration in the current system presents challenges. Manual logbooks are still used for visitor tracking, and contraband detection relies on human oversight, increasing the risk of errors. Additionally, incident reporting and documentation take considerable time, delaying responses to security threats.

Similar challenges exist in other institutions. Many organizations or schools continue to use paper-based logbooks, which become inefficient as visitor numbers grow. Oktaviandri and Foong (2019) highlight that manual logs consume excessive time and often prevent thorough identity verification when managing large volumes of visitors. Furthermore, these records lack long-term traceability, making retrieval difficult after several years.

To address these issues, a Visitor's Log Monitoring System and Contraband Detection System is essential. By integrating digital record-keeping and AI-driven contraband detection, correctional facilities can enhance security, improve efficiency, and ensure a more reliable and transparent monitoring process.

Overview of the Proposed System

The proposed system is designed to make visitor management more efficient and improve security using advanced technology. By integrating image processing tool, it can analyze images to detect contraband items. Since the system operates as an offline website, it remains accessible and keeps data secure even without an internet connection.

Instead of relying on traditional paper logbooks, the system provides a digital platform that automates visitor registration and monitoring. It efficiently records visitor details, visitation schedules, and inmate information, making data retrieval easier while reducing the risk of errors from manual entry. Security staff can use mobile cameras to scan visitor belongings, with the system automatically identifying prohibited items based on a predefined database.

Beyond just detecting contraband, the system helps track security violations by their confiscated items, identifying repeat offenders, and allowing administrators to impose penalties. It also sends real-time alerts when contraband is detected and generates detailed reports on visitor activities and security incidents, which are accessible to both administrators and correctional staff. By replacing outdated manual processes with automation, the system ensures compliance with institutional policies, reduces human error, improves efficiency, and strengthens overall security within correctional facilities.

Hardware and Software Requirements

During the system development phase, acquiring the necessary components is essential for successful implementation. Researchers carefully evaluate and procure both hardware and software to ensure the system operates efficiently without errors or failures. This process involves a detailed assessment of compatible options and consultations with skilled programmers to make informed decisions.

The research team selects hardware based on factors such as compatibility, performance, and scalability to effectively meet system requirements. Similarly, software choices are made with consideration of system architecture to ensure seamless operation. Collaborating with experienced programmers enhances this process, as expertise provides valuable insights into optimal hardware and software configurations, ensuring smooth integration and reliable system performance.

Hardware Requirements

The selection of hardware components is based on compatibility, performance, and scalability, ensuring that the system can handle the demands of real-time visitor monitoring and contraband detection. Devices such as high-resolution cameras, mobile devices, and processing units are evaluated for their ability to support image processing and data storage. Additionally, the network infrastructure must be robust enough to handle secure data transmission and system accessibility within correctional facilities.

Table 3.1 Hardware Requirements

Hardware Component	**Description**	**Quantity**
Administrative Workstation (PC/Laptop)	Manages visitor logs, contraband alerts, and system operations.	1
Server for Database and Application Hosting	Stores and processes visitor records, contraband reports, and security logs.	1
Mobile Devices	Captures and scans images for contraband detection.	2
Network Router	Ensures stable and secure system connectivity.	1
Ethernet Switch	Connects multiple system components via a wired network.	1
Backup Power Supply (UPS)	Prevents data loss during power outages.	1

The preceding Table 3.1 outlines the necessary hardware requirements for the program. The system requires reliable and compatible hardware infrastructure to ensure smooth operation. These components support core functions such as data processing, contraband detection, and secure network connectivity. The following are the required hardware specifications:

a. *Administrative Workstation (PC/Laptop).* Used by administrators to manage visitor logs, monitor contraband alerts, and oversee system operations.
    - CPU. Intel Core i7 or AMD Ryzen 7
    - RAM. 16GB
    - Storage. 1TB SSD + 500GB HDD
    - OS. Windows/Linux
b. *Server for Database and Application Hosting.* Stores and processes visitor records, contraband reports, and security logs.
    - CPU. Intel Xeon or AMD EPYC
    - RAM. 32GB
    - Storage. 2TB SSD with cloud backup integration
    - Network. Gigabit Ethernet
c. Mobile Devices. Used by jail officers to scan visitor belongings via high-resolution cameras.
    - Camera. At least 12MP
    - Processor. Octa-core
    - RAM. 6GB
    - Storage. 128GB

d. *Network Router.* Provides secure and stable connectivity between system components.
   - Type. Enterprise-grade router
   - Network. Supports Wi-Fi 6 and Gigabit Ethernet

e. Ethernet Switch. Ensures stable wired network connections.
   - Ports. 8-port or more Gigabit Ethernet

f. Backup Power Supply (UPS). Prevents data loss and system downtime during power interruptions.
   - Capacity. 1000VA

Software Requirements

The choice of software is equally important, as it must align with the system's architecture and goals. The visitor log monitoring module requires a secure database management system, while the contraband detection feature relies on an image processing tool for accurate analysis. Additional software includes encryption and authentication protocols to ensure data protection and system integrity. Compatibility with the selected hardware and operating system is essential for seamless integration.

An intuitive interface framework enables users to operate the system efficiently, while regular updates help maintain reliability and address security risks. Clear documentation and support tools are also vital for troubleshooting and system maintenance. Moreover, scalability is key to adapting the system to future needs and technological advancements. By selecting appropriate software, the system can function securely and effectively within a correctional facility environment.

Table 3.2 Software Requirements

Software	Description
FRONTEND DEVELOPMENT	
HTML	structures UI elements for accessibility and cross-browser compatibility.
CSS	Styles the UI, enhancing layout, colors, fonts, and responsiveness.
JavaScript	Adds dynamic features like interactive maps and real-time updates for better UX.
BACKEND DEVELOPMENT	
PHP	Handles server-side logic, data processing, heat index calculations, and user authentication.
MySQL	Stores and manages structured data like sensor readings, historical data, and user profiles.
WEB SERVER	
Apache	Hosts PHP scripts and static content, managing HTTP requests from clients.
INTEGRATION AND AI PROCESSING	
Google Cloud Vision API	Analyzes images for contraband detection using AI-powered object recognition.
SECURITY	
HTTPS	Secures communication between clients and the server, protecting sensitive data.
ADDITIONAL TOOLS	
AJAX	Asynchronously fetches data for real-time updates without page reloads.

The preceding Table 3.2 presents a comprehensive list of the software tools, technologies, and platforms essential for the development, deployment, and operation of the system. This includes both frontend and backend components, as well as web server configurations, integration services, and security protocols. Each software item is carefully selected to fulfill a specific role in ensuring the system’s overall functionality, efficiency, and security.

The frontend tools are designed to provide a user-friendly interface for administrators and correctional officers, while the backend components manage database interactions, contraband detection, and other system processes. Additionally, web server configurations enable seamless communication between the system's various components, ensuring smooth data flow and real-time updates.

The integration services ensure that the system works cohesively with existing infrastructure, while security protocols protect sensitive information and prevent unauthorized access. Each element plays a critical part in optimizing performance and safeguarding the integrity of the system. The table serves as a reference for understanding the technological foundation of the project, outlining the necessary software to create a robust and reliable solution for correctional facility management.

CHAPTER IV

DESIGN AND METHODOLOGY

This chapter examines the numerous methodology and research methods that information systems researchers often utilize. Design and methodology were used to plan a study and provide an overall framework for collecting data. It established how an investigation should proceed. This study's methodology and approach are acknowledged.

Waterfall Diagram

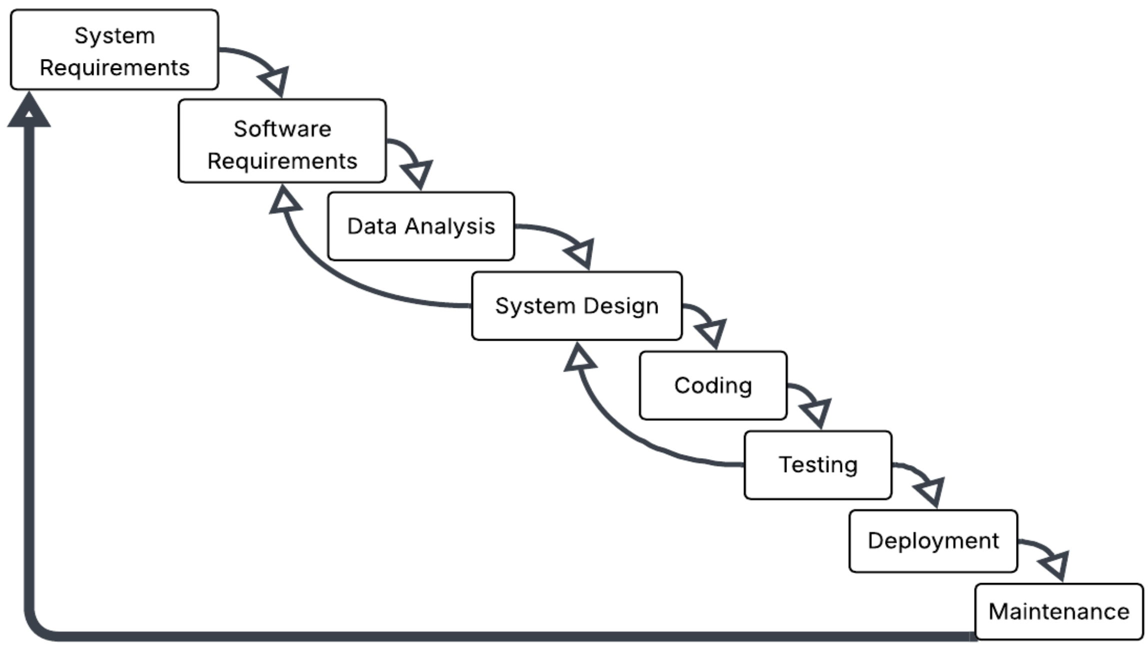

Diagram 4.1 Waterfall Diagram

The Diagram 4.1 above illustrates the stages and processes followed in conducting this study using the Waterfall Model. The methodology consists of sequential phases, where each phase must be completed before proceeding to the next. If issues arise, the process returns to the previous phase for necessary modifications. This structured approach ensures that each phase is thoroughly reviewed and validated before moving forward, minimizing errors and enhancing the quality of the final output.

System Requirements

The implementation of the Visitor's Log Monitoring System and Contraband Detection in Correctional Institutions necessitates specific hardware components to ensure optimal functionality and efficiency. Surveillance cameras are required to monitor visitor activities and capture images for contraband detection, while a mobile device with a camera facilitates the scanning of visitor belongings using Google Cloud Vision API for image analysis.

A dedicated server or workstation is essential for processing and storing visitor logs, contraband reports, and other critical system data. Additionally, networking equipment, such as routers and switches, is necessary to enable seamless communication between system components. To support data management and ensure long-term storage, high-capacity storage devices are required to securely maintain images, visitor records, and reports. The integration of these hardware components is crucial in ensuring the system's reliability and operational efficiency.

Software Requirements

The software infrastructure of the system incorporates advanced technologies to support its core functionalities. Google Cloud Vision API is utilized for contraband detection through image processing, enhancing security by identifying prohibited items during visitor inspections. The web-based platform is developed using HTML, CSS, and JavaScript for the user interface, while a backend framework such as PHP manages system operations and data processing. A Database Management System (DBMS), such as MySQL is implemented to efficiently store visitor logs, scanned images, and generated reports. To maintain offline accessibility, the system is hosted on a local web server, such as Apache. Additionally, the system operates on a secure operating system, such as Windows Server or Linux, ensuring stability and compatibility with system requirements. The combination of these software components establishes a robust and secure platform tailored to the system's monitoring and security requirements.

Data Analysis

In this phase, the researchers ensured data accuracy, reliability, and system functionality. The Waterfall Model was used due to its step-by-step structure, supporting a clear sequence for system development. The process started with a review of correctional facility policies to understand visitation rules and contraband restrictions. Existing systems for visitor management and contraband detection were compared to identify security gaps, as shown in Table 4.1. Surveys and interviews with correctional officers and staff provided insights that helped refine system requirements.

The data collected was then organized into a clear format for effective use in system design. A flowchart, presented in the succeeding Diagram 4.2, was also created to illustrate the current manual procedures for visitor registration, inspection, contraband detection, and handling violations. This helped pinpoint inefficiencies, security concerns, and opportunities for automation.

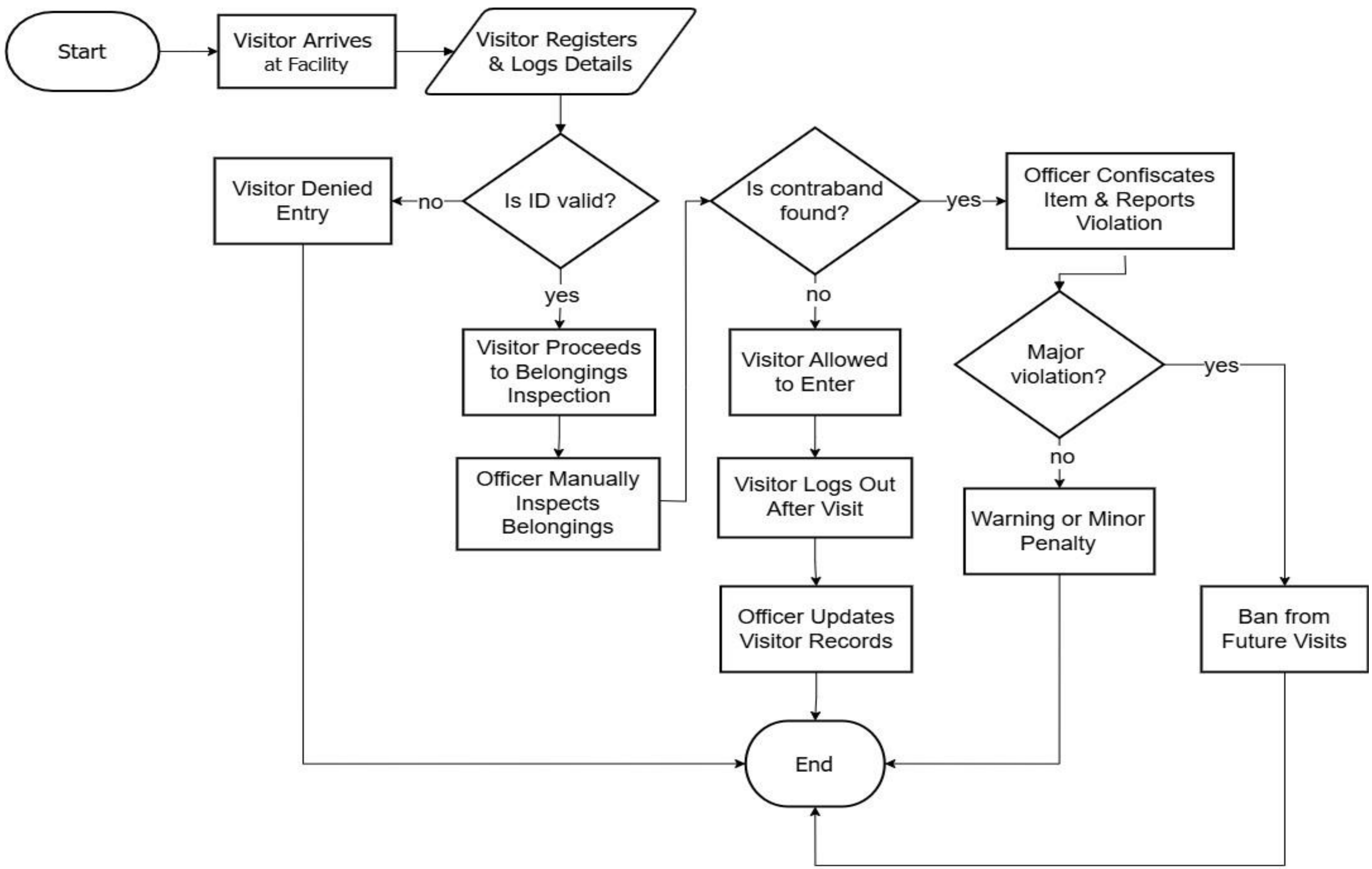

Diagram 4.2. Flowchart

The process depicted in Diagram 4.2 precedingly illustrates the step-by-step manual procedure of visitor registration, belongings inspection, and contraband detection within the correctional facility. It visually represents how visitors log their details, how correctional officers inspect visitor belongings, and the decision-making process when contraband is detected, including confiscation, reporting, and possible visitor bans. This flowchart helps identify inefficiencies in the manual process, highlighting areas where automation and AI-powered detection can improve security, accuracy, and efficiency.

Comparative Analysis Table (Existing vs. Proposed System)

To better understand the benefits of transitioning to a modernized digital system, the following table outlines a side-by-side comparison between the current manual processes and the proposed automated solution:

Table 4.1. Comparative Analysis

Aspect	**Existing System**	**Proposed System**
Visitor logging	Uses manual logbooks, prone to human error.	Utilizes automated digital logging with secure database storage.
Contraband detection	Relies on manual inspection.	Implements AI-powered image analysis using Google Cloud Vision API.
Security measures	Based on paper records, vulnerable to manipulation.	Implements role-based access control and encrypted records.
Efficiency	Time-consuming manual checks.	Offers faster, automated detection and notification.

The preceding Table 4.1 highlights the key differences between the existing manual system and the proposed Visitor's Log Monitoring System and Contraband Detection in Correctional Institutions. It focuses on critical aspects such as visitor logging, contraband detection, security measures, and efficiency. The table clearly illustrates how the proposed system enhances accuracy, automates processes, strengthens security, and improves overall operational efficiency compared to the traditional manual approach.

By ensuring compliance with institutional regulations while integrating modern technologies, this system will optimize operational efficiency, minimize human error, and enhance security measures. The insights gained from this phase serve as the foundation for the next stages of development, ensuring that the proposed solution effectively addresses the identified challenges.

System Design

The system design phase involves translating the gathered requirements into a structured architectural plan. This stage defines the system's overall structure, components, and interaction flow. The database design was developed to store visitor logs, timestamps, contraband detection reports, and security alerts. The system architecture was designed to consist of a front-end interface for visitor logging, a back-end processing module for contraband detection, and an API integration with Google Cloud Vision for image-based scanning. The user interface (UI) was carefully designed to provide an intuitive and efficient experience for correctional staff, featuring real-time visitor check-ins, contraband alerts, and an interactive dashboard for monitoring security data. If any major issues arise during the later stages of development, adjustments to the system design are made before proceeding further.

Coding

Once the system design was finalized, the actual development process began. The coding phase involved translating the system specifications into a functional software application. This stage required the implementation of visitor logging functionalities, where check-in and check-out records were accurately recorded and stored in a secure database. The contraband detection module was developed using image processing technology, enabling the system to analyze images and flag prohibited items. Security features such as role-based authentication were integrated to restrict access based on user permissions, ensuring that only authorized personnel could manage visitor records and contraband reports. During this phase, iterative debugging and code optimization were conducted to identify and resolve errors. If critical issues or inefficiencies were encountered, the process would return to the system design phase for necessary refinements.

Testing

Testing is a crucial phase to ensure that the system operates correctly and meets the defined requirements. Multiple testing methodologies were used, starting with functional testing, which verified that the visitor logging system correctly recorded check-in and check-out data and that the contraband detection module accurately flagged prohibited items. Security testing was conducted to assess the system’s resistance to unauthorized access, data breaches, and potential cyber threats. Performance testing evaluated the system's speed, scalability, and responsiveness under high-volume visitor entries. Additionally, usability testing involved correctional staff using the system in a controlled environment to ensure the interface was intuitive and easy to navigate. If significant issues were identified during testing, the development process returned to the coding phase for corrections before proceeding to deployment.

Deployment

After successful testing, the system was deployed for real-world use in the correctional institution. This phase involved installing the software on authorized devices and ensuring seamless integration with existing security infrastructure. Training sessions were conducted for correctional staff to familiarize them with the system's functionalities, such as logging visitors, reviewing contraband alerts, and managing security reports. Security protocols were reinforced to prevent unauthorized access, and initial monitoring was performed to identify any post deployment issues. The deployment phase also included live testing in an operational environment, allowing real-time visitor data collection and contraband detection. If unexpected issues arose during implementation, modifications were made by revisiting the coding or testing phases before full-scale use.

Maintenance

The final phase involves the ongoing maintenance and improvement of the system to ensure its long-term efficiency and reliability. Regular monitoring is conducted to track system performance, detect security vulnerabilities, and address any software bugs. Updates to the contraband detection module are periodically implemented to improve recognition accuracy and expand the database of prohibited items. Feedback from correctional staff is collected to identify areas where usability can be enhanced. Security features are continuously updated to protect against evolving cyber threats, ensuring compliance with the latest correctional facility regulations. If major updates are required, the process loops back to the requirements gathering phase to redefine system needs before implementing enhancements.

Gantt Chart

The Gantt chart for the proposed system development is structured into five distinct phases to guide the project from start to finish. The first phase, Planning and Development, involves laying the foundation for the project by defining objectives, identifying key tasks, and outlining resources. This phase ensures the project is well-organized and sets clear milestones.

Following this, the Data Analysis and Finalization phase focuses on organizing and finalizing the required data for the system, ensuring that accurate and clean data is available for the system's functions. The third phase, Designing, centers on developing the system architecture and user interface, ensuring the system is intuitive and meets the needs of its users, such as administrators and staff.

The Implementation phase is where the system's development begins, including the creation of the backend infrastructure and integration of key technologies like AI-based contraband detection. After initial testing, the Revision phase allows for refinements, addressing issues found during testing and gathering feedback from users to optimize performance. Finally, in the Finalization phase, the system undergoes final testing, and any remaining issues are addressed. The system is then deployed, and training for administrators and staff is conducted.

Legend:

1-week

½week

Task	March				April	
	Week 1	2	3	4	Week 1	2
Brainstorming						
Title identification						
Gathering information						
Creation of chapter I for title defense						
Creation of ppt from chapter I						

Chart 4.1 Gantt Chart (Phase I)

The Chart 4.1 above illustrates the collaborative efforts of the researchers as they coordinated, strategized, and brainstormed to develop the system. This process entailed the exchange of ideas and collective decision making among team members to finalize the project title. Collaborative activities, including title selection and information gathering and then, we completed the creation of Chapter I for the title defense within half a week. Subsequently, we finished the creation of the PowerPoint presentation and prototype within one week.

Phase II (Data Analysis and Finalization)

Task	May		June	
	Week 1	2	Week 1	2
System objectives				
Identify limitations				
Making chapter I of the approved title				
Conducted interview				
Finalized chapter I- III				
Group meeting				

Chart 4.2 Gantt Chart (Phase II)

As presented in Chart 4.2 above, it illustrates a systematic progression of tasks aimed at fulfilling the research objectives. Within the first half-week, the team completed Chapter I, laying the groundwork for the study. Over the following week, the team conducted interviews to obtain data and finalized Chapters II and III, ensuring comprehensive analysis. Regular group meetings were held to enhance communication and alignment on project milestones.

These sessions allowed real-time adjustments and helped address challenges, ensuring timely revisions and improving overall thesis quality. This phase also involved coordinating feedback from the adviser and incorporating suggested revisions. All tasks were carefully scheduled to maintain steady progress and avoid delays.

Phase III (Designing)

Task	June		July			
	Week 3	4	Week 1	2	3	4
Waterfall Diagram						
Gantt Chart						
Context Diagram						
Data Flow Diagram (level 1)						
Use Case Diagram						
Activity Diagram						
Hierarchy Diagram						
Group Meeting						

Chart 4.3 Gantt Chart (Phase III)

Chart 4.3 above illustrates the scheduled development of various system components, each with a defined timeframe. The waterfall diagram begins in the third week of June and spans the entire week. The Gantt chart is set for completion in the fourth week of June, followed by the Data Flow Diagram (Level 0 and Level 1) in the same period. The Use Case Diagram is scheduled for the first week of July, while both the hierarchy and activity diagrams are planned for the second and third weeks of July. Additionally, group meetings to review progress and discuss necessary adjustments are scheduled for the latter part of the second week of July, as well as the fourth weeks of both June and July. These collaborative sessions help maintain alignment and support timely completion of each design phase. Each component's schedule reflects its role in shaping the overall system structure.

Phase IV (IMPLEMENTATION)

Task	August				September			
	Week 1	2	3	4	Week 1	2	3	4
Implementation Planning								
System Design								
System Definition								

Chart 4.4 Gantt Chart (Phase IV)

Refer to Chart 4.4 above, which marks the start of the coding phase. A two-month period was allocated for implementation planning, spanning the first to the second weeks of August. System design followed from the third week of August to mid-September, then system definition was conducted during the third and fourth weeks of September. This structured schedule supported steady progress and a solid foundation for development.

Phase V (Revision and Implementation)

Task	November		December		January	
	Week 1	2	Week 1	2	Week 1	2
System Implementation						
Revision						
Presentation						

Chart 4.5 Gantt Chart (Phase V)

As presented in the preceding Chart 4.5, the researchers engage in the system implementation process, spanning the first and second week of October. Subsequently, revisions to the study and adjustments to the system were conducted throughout the first and second week of November. These endeavors persisted until the first half week of December, dedicated to preparing the final presentation of the study, slated for delivery during the second week of December.

Context Diagram

The Context Diagram provides a high-level visual representation of the interactions between the Visitor’s Log Monitoring and Contraband Detection System and its external entities. It illustrates how data flows between the system and users such as administrators, jail officers, and visitors. The diagram identifies major inputs like visitor registration details, contraband image data, and system credentials, as well as outputs such as alerts, reports, and access logs. It highlights how users submit data, receive feedback, and interact with system processes, including identity verification and item scanning. By mapping these external exchanges, the Context Diagram ensures a clear understanding of the system’s boundaries, communication channels, and operational scope. This overview helps identify integration points, data sources, and potential improvements to ensure effective, secure, and streamlined system use.

As shown in the subsequent DFD (Diagram 4.3) bellow, the Visitor’s Log Monitoring System and Contraband Detection in Correctional Institutions provides an overview of how visitor information is processed, contraband is detected, and security protocols are enforced. The system involves four primary entities: Admin user (managed by Admin), Visitor, and Mobile Camera. Upon arrival, personal details such as name, identification, and purpose of visit must provide by the visitors, which are manually recorded by staff into the system. Visitors do not have access to log in or create accounts within the system. The details are stored for tracking and security purposes. The user or staff, managed by the admin is responsible for handling visitor records, verifying provided details, and monitoring contraband alerts. During the security screening, a mobile camera integrated with image processing tool is used to scan visitor belongings. The system processes the images and detects any prohibited items based on predefined contraband categories. If a restricted item is identified, the system generates an alert, prompting staff to take necessary action, such as confiscation, denial of entry, or further security checks.

The admin has full control over the system, managing authorized users such as correctional staff or guards. Admin are responsible for creating and assigning user roles, overseeing visitor logs, updating contraband detection, settings, and reviewing security reports. These reports help analyze trends in contraband incidents and visitor activity, allowing the administration to refine security policies and improve overall system efficiency.

By digitizing visitor logs and integrating AI-powered contraband detection, the system enhances security monitoring, reduces human error, and ensures a more efficient and transparent visitor screening process within the correctional facility. It also maintains accurate, easily retrievable records while minimizing manual workload for correctional staff.

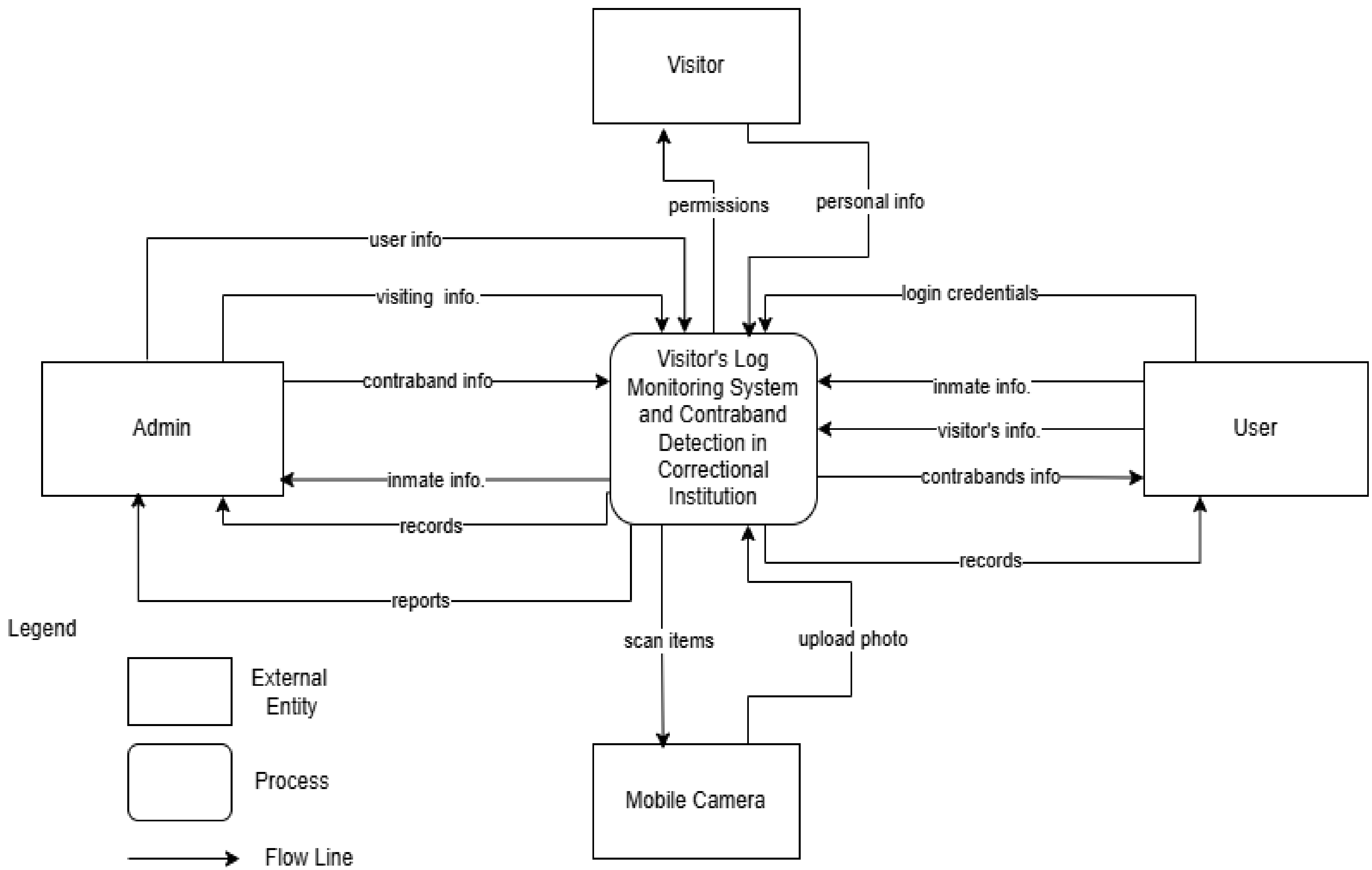

Diagram 4.3 Context Diagram

Data Flow Diagram Level 1

The Reference Data Flow Diagram (DFD) Level 1 provides a detailed breakdown of the system’s key processes and data interactions. It builds upon the Level 0 DFD by illustrating sub-processes such as visitor registration, identity verification, contraband detection using an image processing tool, and system monitoring. This level clarifies how information moves between data stores and external entities.

The succeeding Diagram 4.4 illustrates how information flows within the BJMP system between external entities (Admin and Staff), processes, and data stores. The admin manages guard schedules, staff accounts, inmates, contraband definitions, visitor banning, and visiting hours, while Staff focuses on inmate management, visitor registration, and monitoring. Key processes include Authentication (1.0), which verifies credentials against tbl_bjmp_admin and tbl_users; Inmate Management (5.0) for maintaining inmate records; Visitor Registration (8.0) for processing visitors and creating visit records; Contraband Checking (9.0) for scanning visitors against defined contraband items; and Visitor Monitoring (10.0) for tracking active visits using rules from tbl_visiting_hours and tbl_visiting_rules. The system utilizes 14 database tables as data stores, with labeled flow lines showing data movement between components (e.g., "login credentials," "save," "fetch"). This diagram effectively captures the system's operational flow from user input through processing to data storage and retrieval, demonstrating how the integrated components work together to manage the facility's operations.

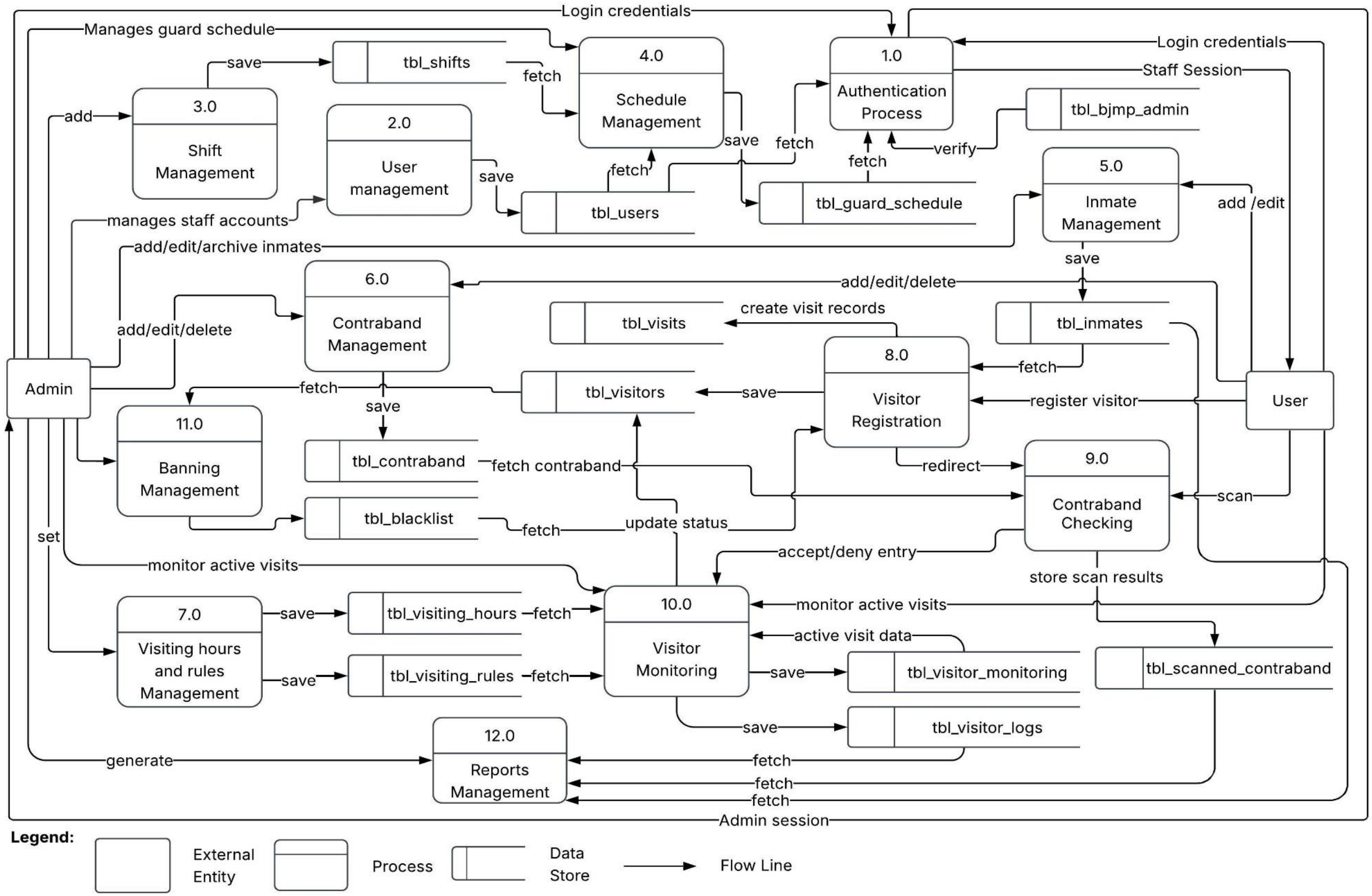

Diagram 4.4 Data Flow Diagram (Level 1)

Use Case Diagram

The Use Case Diagram for the proposed Visitor's Log Monitoring and Contraband Detection System provides a visual representation of how various users interact with the system within a correctional facility. It highlights two main user roles: administrators and jail officers, showing the specific tasks each role is responsible for. Administrators handle visitor information, manage user access, and generate system reports, while jail officers conduct contraband checks using AI-assisted image analysis.

This diagram is important for clarifying user responsibilities and system boundaries. It outlines which features are accessible to each role and shows how alerts, logs, and notifications are processed. By mapping these interactions, it helps maintain secure access control and supports system efficiency. It also identifies opportunities for improvement by analyzing user interactions with features such as real-time alerts and automated detection. This understanding serves as a foundation for future updates, ensuring the system adapts to evolving security needs in correctional facilities

The succeeding Diagram 4.5 presents the system flow, starting with user login. Users play a key role in adding inmate and visitor records, managing visitor logs, updating or deleting entries, and reporting incidents such as contraband discovery. The system also supports printing reports based on logs and analysis results, improving documentation and supporting real-time monitoring. It ensures that tasks are securely tracked and assigned to appropriate roles. Finally, both admin and users log out to maintain data integrity and security.

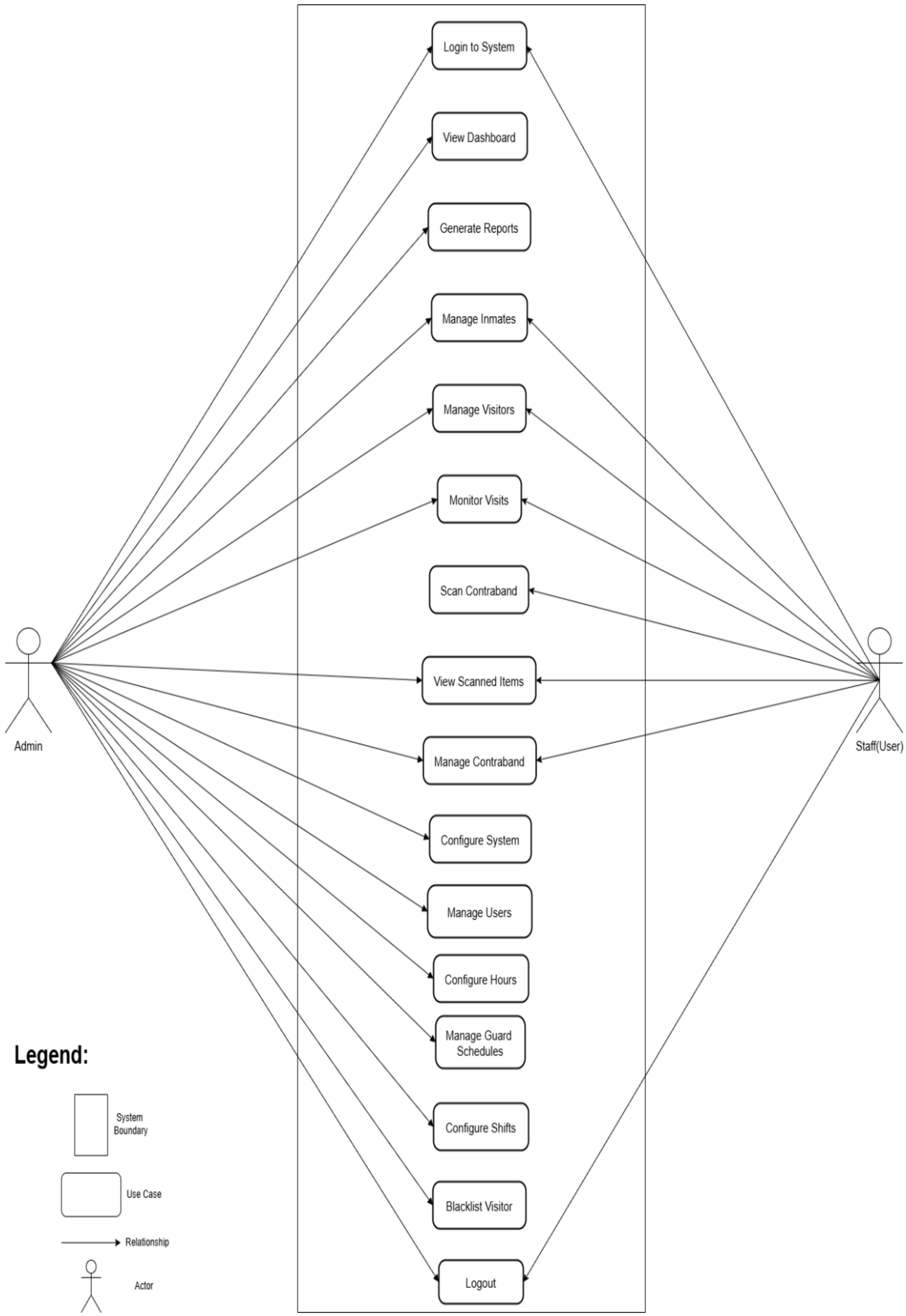

Diagram 4.5 Use Case Diagram

Activity Diagram

The Activity Diagram illustrates the step-by-step workflow of the Visitor's Log Monitoring System and Contraband Detection in a Correctional Institution. It visually represents the sequence of activities performed by different users, including the admin and staff. The process begins when a visitor provides personal details, followed by staff verifying and logging the entry. If items are brought for visitation, it is being scanned using a mobile camera for contraband detection. If contraband is detected, an alert is generated, and staff decide whether to deny entry or confiscate the item. The system then updates the visitor log and security reports. This diagram helps stakeholders understand user interactions and how decisions are made at each stage of the process.

The succeeding Diagram 4.6 illustrates the activity flow for the BJMP system, organized into three swim lanes: Admin, System, and Staff (User). The admin workflow begins with receiving a visitor and verifying the corresponding information, followed by a decision point for contraband checking, which leads to either approval or denial. The System Lane handles the core processing, including validating data, checking schedule records, notifying users, and managing time tracking with appropriate validations at key decision points. The Staff workflow involves visitor arrival, verification, contraband checking, and registration, with decision points for contraband detection and visitor approval. The diagram effectively shows how activities flow between actors, with clear decision points determining alternative paths and endpoints, demonstrating the complete visitor processing sequence from initial contact through final approval or denial.

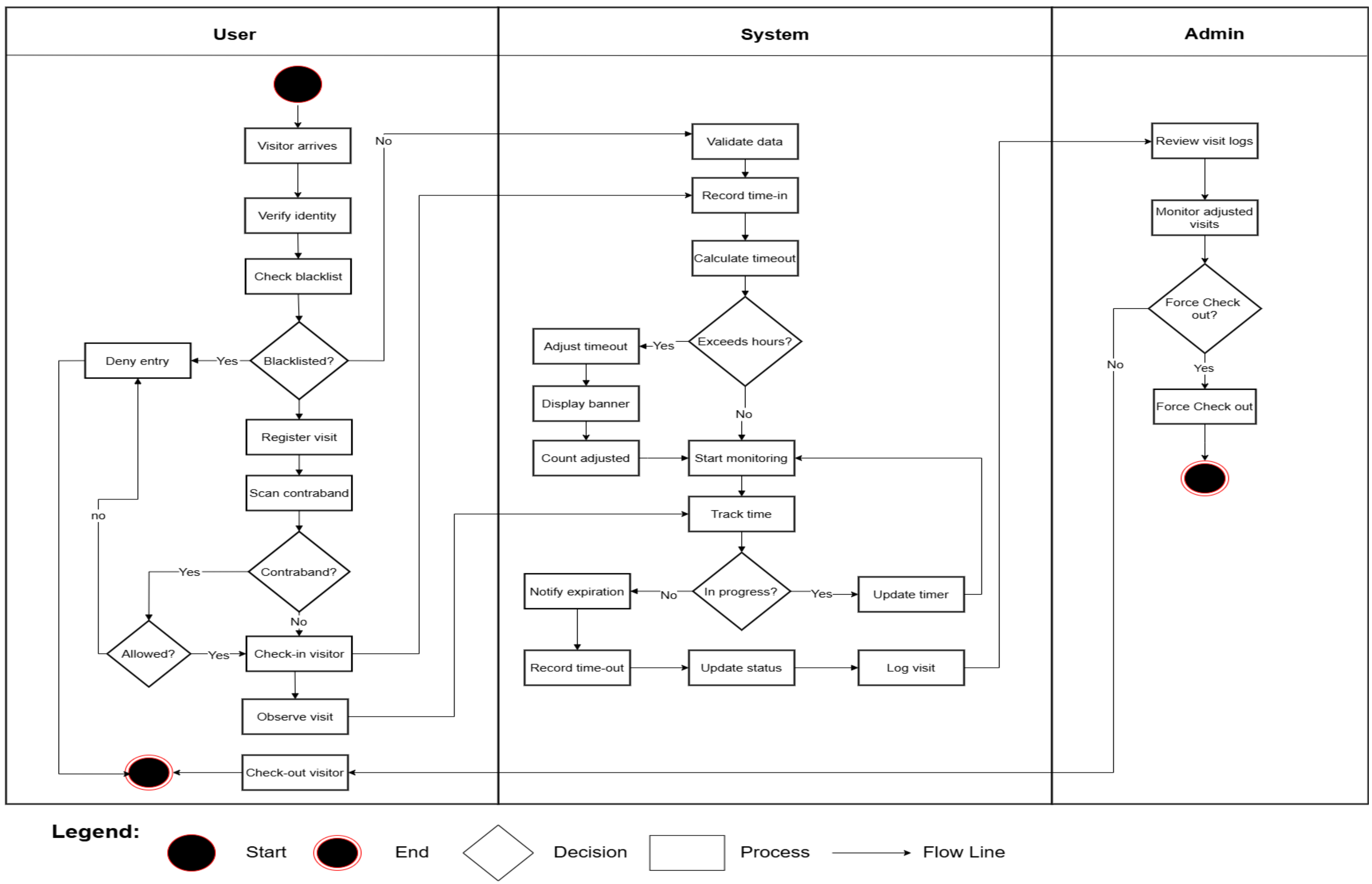

Diagram 4.6 Activity Diagram

Hierarchy Input Process Output (HIPO) Diagram (Admin)

The Hierarchy Input Process Output (HIPO) diagram provides a structured overview of the system's main functions and their relationships. It outlines how inputs, processes, and outputs are organized to support core tasks like visitor logging and contraband detection. This helps clarify the system's overall flow and structure while serving as a guide for more detailed diagrams and development phases. It also aids in visualizing how each module contributes to the overall functionality of the system.

The succeeding Diagram 4.7 illustrates the admin's hierarchical structure within the BJMP system. The diagram begins with the Login function, which serves as the entry point for all administrative activities. From this central access point, the admin can navigate to five key management areas. First, the Admin manages Inmate Information, allowing them to maintain comprehensive inmate records through the Manages Inmates Records process, resulting in organized Inmates Records. Second, they oversee Contraband Information, enabling them to define and update prohibited items via the Manage Contraband process, which produces detailed Contraband Records. Third, the Admin handles User Information, where they can create and modify staff accounts through the Manage User Accounts process, generating User Account Records. Fourth, they control Schedule Parameters, setting up and adjusting staff work shifts through the Set User Schedule process, which creates User Schedule Assignments. Finally, they configure Visiting Hours/Rules Information, defining visitation policies through the Define Visiting Rules process, resulting in Visiting Hours Configuration.

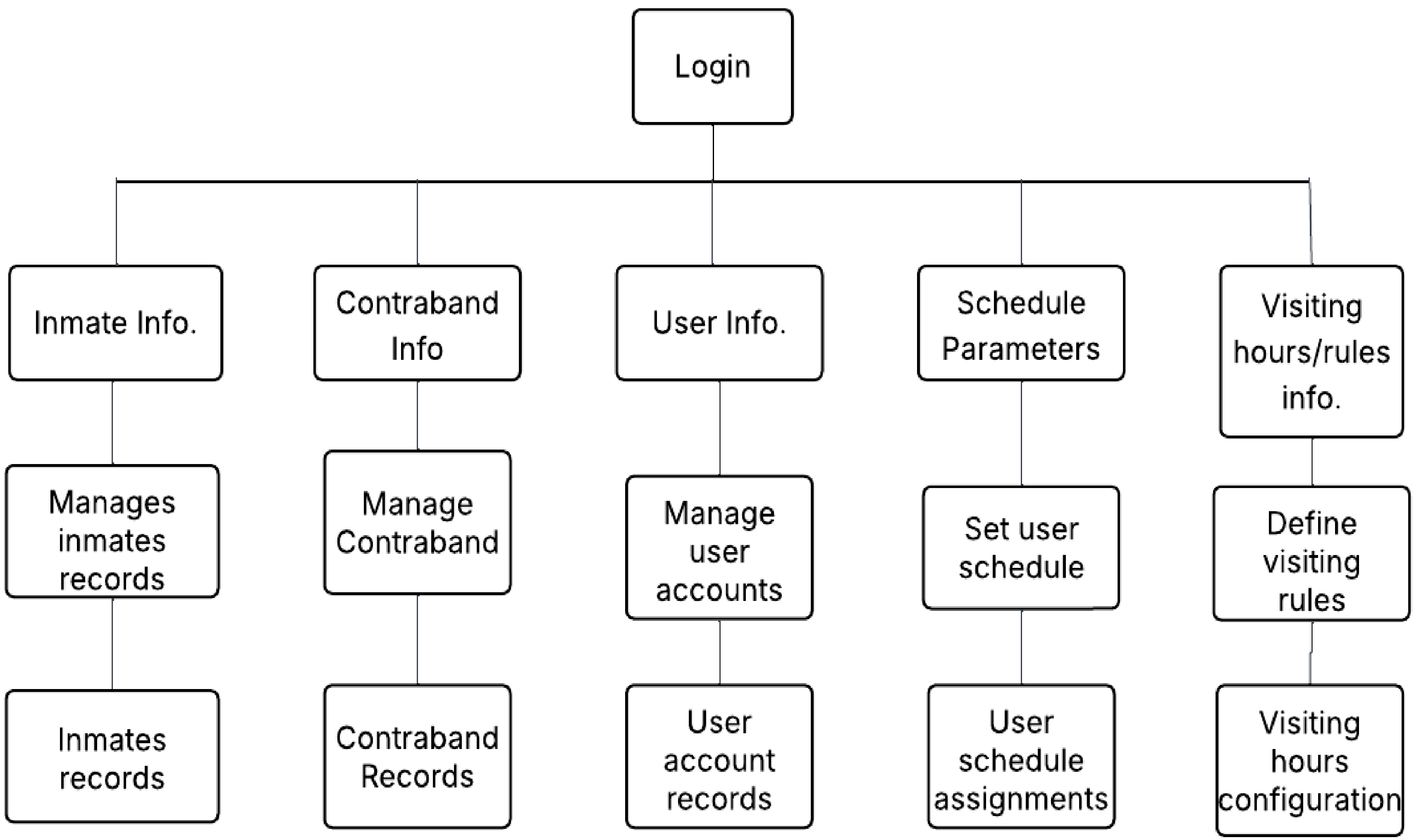

Diagram 4.7 Hierarchy Diagram (Admin)

Hierarchy Input Process Output (HIPO) Diagram for User

The HIPO diagram outlines the Staff's key responsibilities in handling visitor logs, monitoring for contraband, and enforcing security protocols. Inputs include visitor information, scanned data from belongings, and system-generated alerts. Processes involve logging entries, verifying items, detecting contraband using image analysis, and recording any violations. Outputs include access approvals or denials, contraband alerts, and incident reports. This structure promotes efficient visitor management while ensuring adherence to the facility's security policies. It also provides a clear overview of how data flows through the system and how responsibilities are divided among staff.

As shown in the succeeding Diagram 4.8, it illustrates the Staff's hierarchical structure in the BJMP system. Starting with Login as the entry point, the Staff manages four key operational areas. First, they handle Inmate Information, allowing them to manage inmate records and maintain comprehensive inmate documentation. Second, they process Visitor Information and Photos, enabling them to manage contraband screening and generate contraband records for each visitor. Third, they utilize Contraband Scan Images to scan and detect prohibited items, producing detailed contraband detection results. Fourth, they oversee Contraband Information, which allows them to manage contraband databases and generate contraband results reports. This structured hierarchy ensures efficient visitor processing, thorough security screening, and proper documentation of all contraband-related activities within the facility.

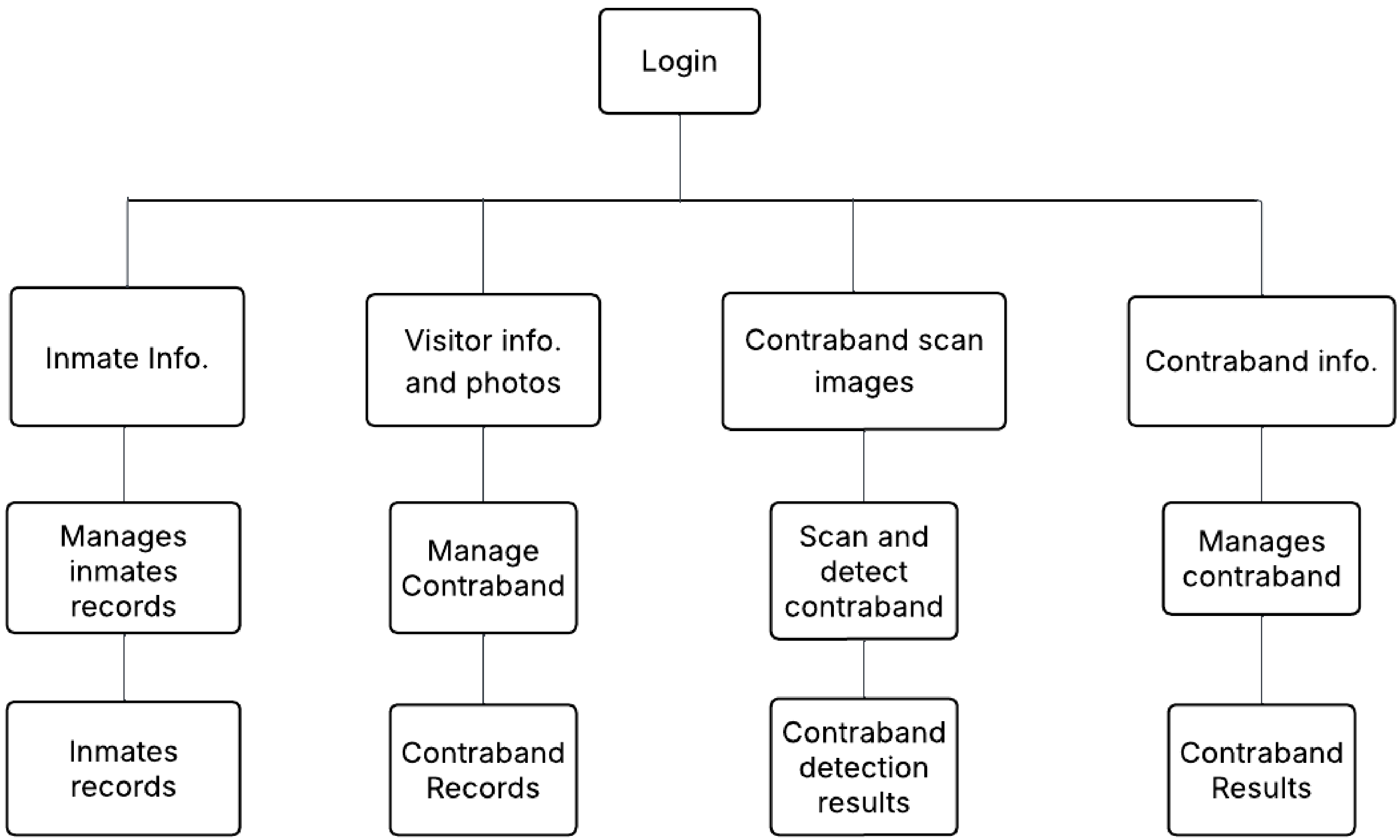

Diagram 4.8 Hierarchy Diagram (User)

Entity-Relationship Diagram (ERD)

The Entity-Relationship Diagram (ERD) defines the structure of the system, illustrating key entities and their relationships. The admin manages staff accounts, visitor records, inmate details, contraband reports and handle violations. Staff/Guards log visitor details, monitor scanned items, Visitors provide identification upon entry, while Inmates are linked to visitation records. Contraband Detection processes scanned images and generates alerts for prohibited items.

As shown in the succeeding Diagram 4.9, it depicts the database structure for the Visitor's Log Monitoring System and Contraband Detection in Correctional Institution. The diagram shows 14 interconnected tables that support inmate visitation management, security operations, and staff scheduling. The central tables include tbl_visitors (visitor information), tbl_inmates (inmate data), tbl_visits (visitation records), and tbl_visitor_monitoring (real-time visitor tracking). Security is managed through tbl_contraband and tbl_scanned_contraband tables, while system users are represented by tbl_bjmp_admin and tbl_users tables. Configuration tables include tbl_visiting_hours, tbl_visiting_rules, tbl_shifts, and tbl_guard_schedule, which define operational parameters.

The relationships between tables (shown as connecting lines) establish how visitors interact with inmates, how staff monitor these interactions, and how the system enforces security protocols. This ERD effectively illustrates the database architecture that enables visitor processing, inmate management, contraband detection, and staff scheduling within correctional facilities.

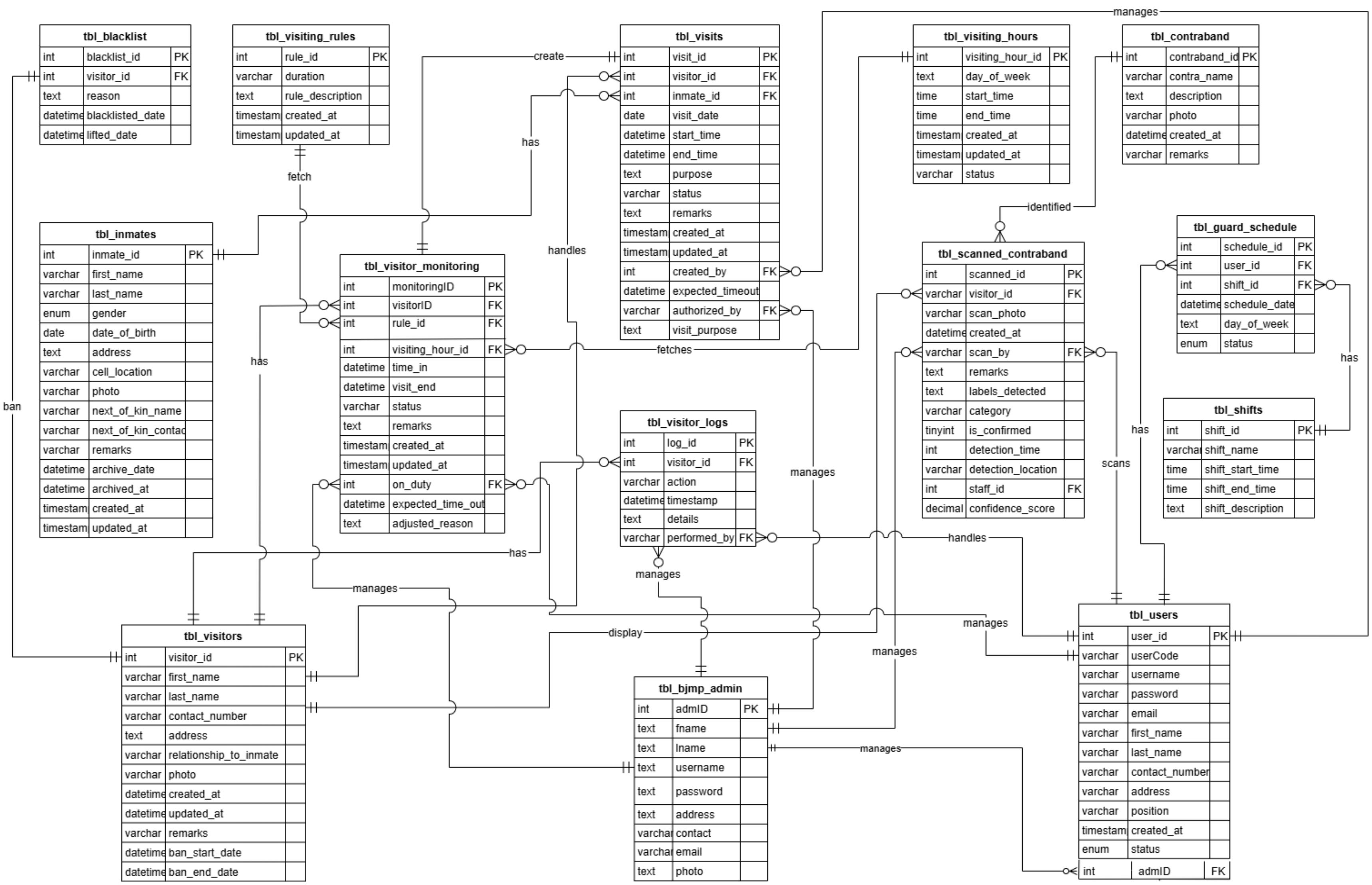

Diagram 4.9 Entity Relationship Diagram

Graphical User Design

A Graphical User Interface (GUI) is a visual design framework that allows users to interact with a system through elements like icons, buttons, and windows, instead of relying on text-based commands. Its primary purpose is to simplify system use by making it more intuitive and accessible, especially for users who may not have technical expertise.

GUIs enhance user experience by providing clear visual feedback, reducing the learning curve, and making complex tasks easier to perform. In the context of a study or system design, such as a prison management system, a GUI plays a vital role in displaying how users will interact with the system. It helps present interface layouts as visual guides, making it easier to demonstrate the system's functionality, gather user feedback, and improve usability. Including GUI designs in research or documentation also enhances clarity and makes it easier to communicate the system's features to stakeholders, contributing to more effective development and evaluation.

Admin

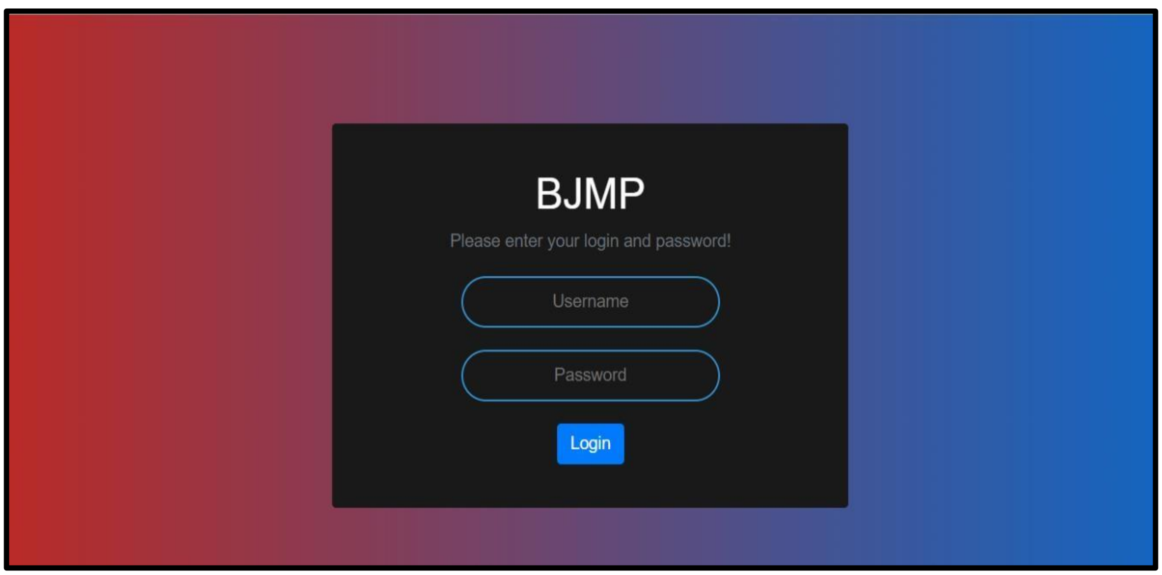

Figure 1. Log in interface of the admin

As illustrated in the preceding Figure 1, the admin login interface comprises a username field, password field, login button. To gain access to, the system (Bureau of Jail Management and Penology). input the unique username and password into the respective fields and subsequently click the login button.

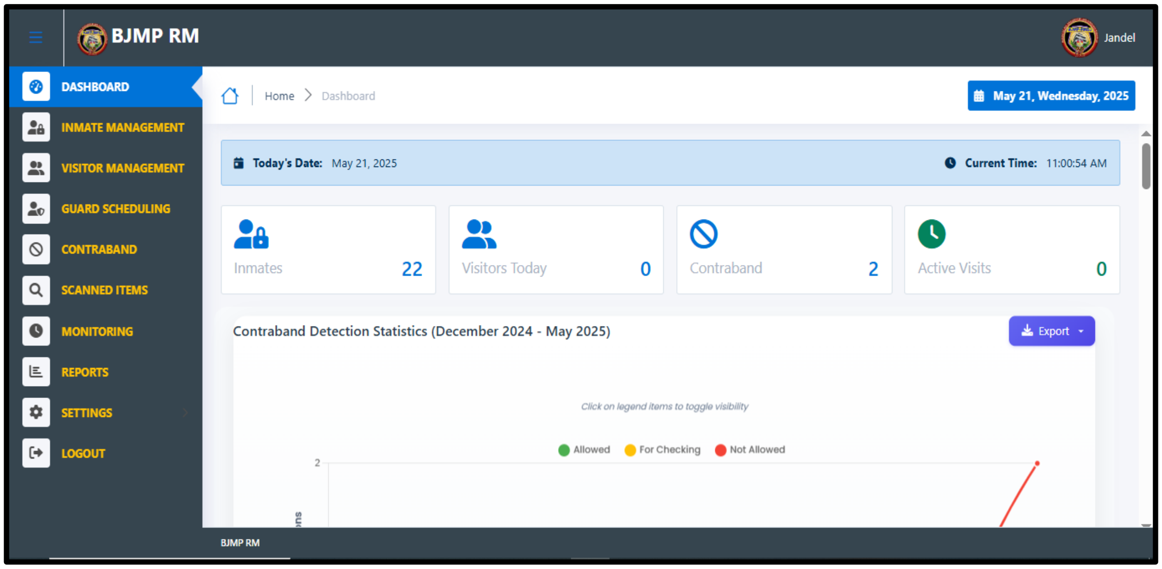

Figure 2. Dashboard of the admin

In Figure 2 above, shows the dashboard for a prison management system. The dashboard provides a snapshot of key metrics, including the number of inmates, visitors today, contraband item detected and statistical reports.

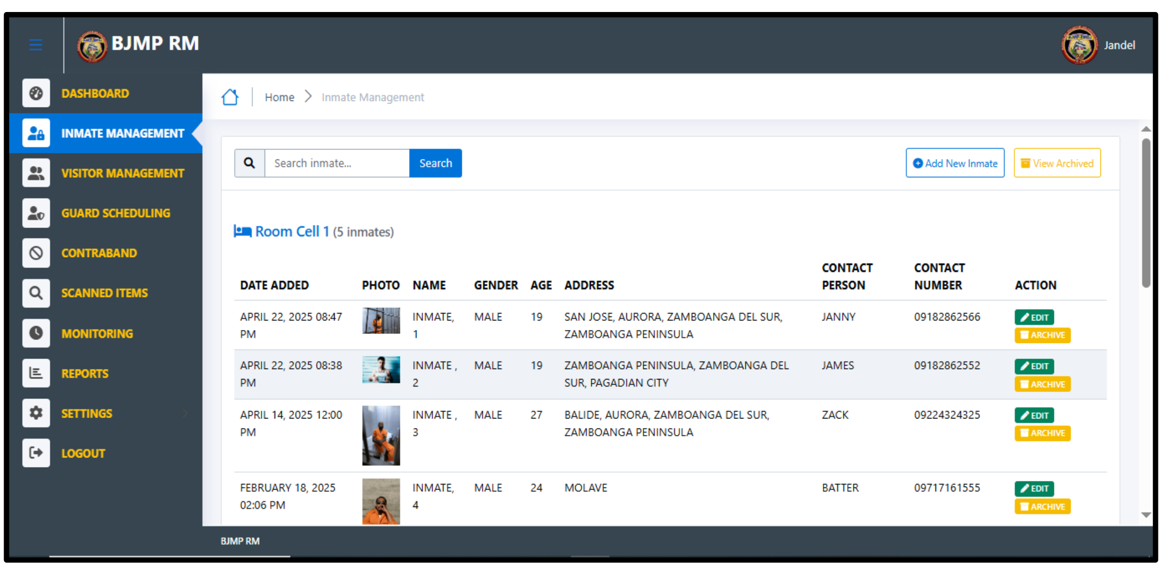

Figure 3. Inmate Management

The preceding Figure 3 displays inmates organized by room cells with comprehensive details including photos, personal information, and contact details. The interface features search functionality and action buttons for record management. The left navigation panel provides access to all system modules, enabling administrators to efficiently manage inmate records while maintaining a clear overview of facility occupancy by cell location.

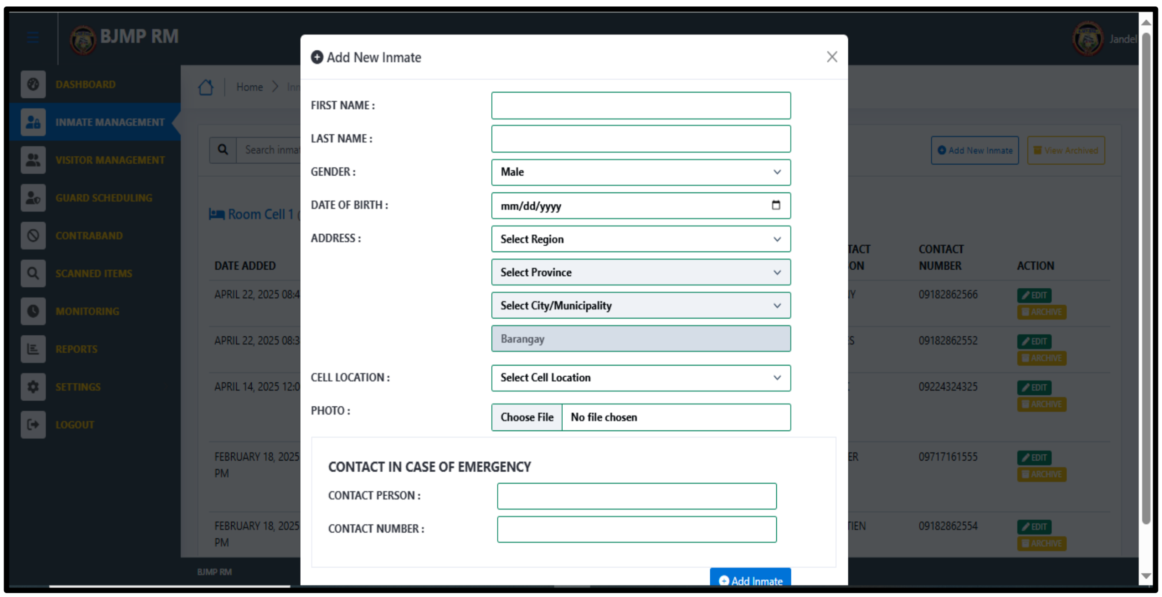

Figure 4: Add Inmate

The above Figure 4 shows the inmate adding form in the system, displayed as a modal overlay on the inmate management interface. It includes fields for key details such as full name, age, gender, and birthdate, along with dropdown menus for address selection to ensure consistency.

The form also allows cell location assignment, photo upload for visual ID, and emergency contact entry. These elements streamline the intake process, improve data accuracy, and support efficient management. A submit button finalizes the entry and saves the data in the system.

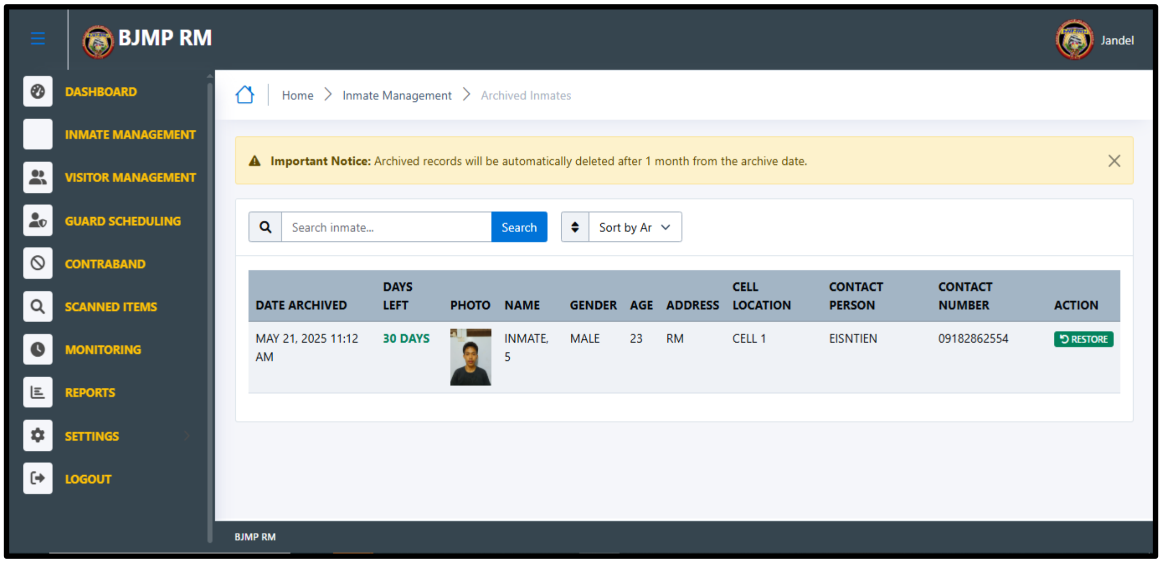

Figure 5. Archived Inmate

The Figure 5 above displays the Archived Inmates interface showing temporarily removed inmate records with their archive date, days left before permanent deletion, and complete inmate information. A warning notice indicates that archived records will be automatically deleted after one month. Each record includes a restore button allowing administrators to reinstate inmates to the active roster when needed.

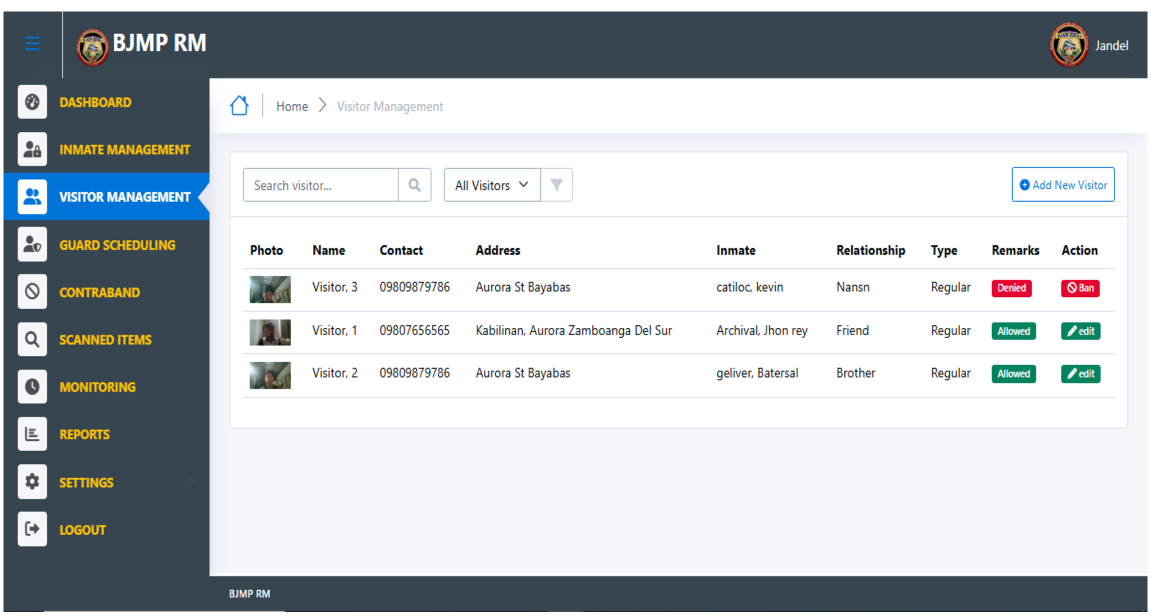

Figure 6. Visitor Management

The preceeding Figure 6 shows a visitor management system. The interface displays a list of visitors, with columns for their photo, name, address, the inmate they are visiting, their relationship to the inmate, the type of their visit, and actions that can be taken (editing, or deleting the record).

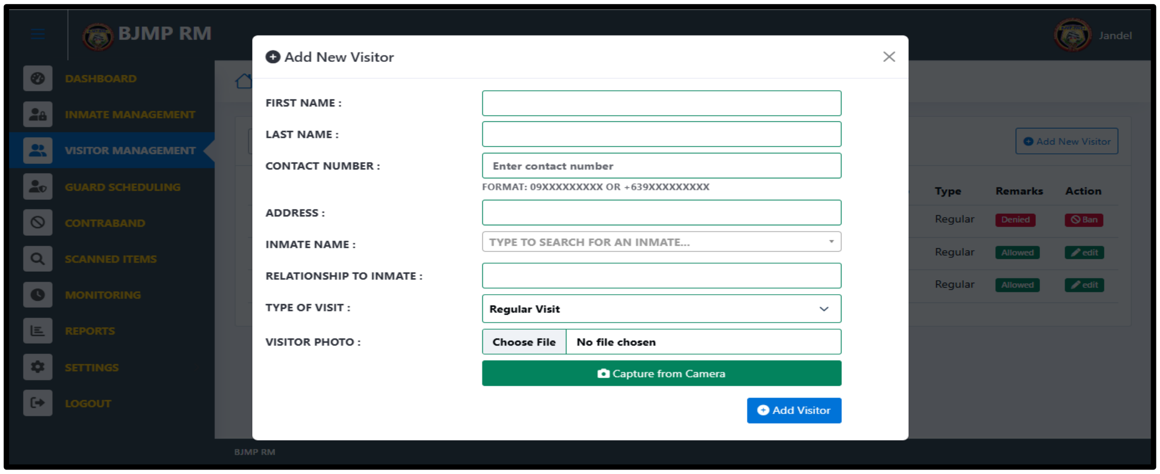

Figure 7. Add Visitor

The Figure 7 above shows the visitor registration form used to collect essential details such as the visitor's name, relationship to the inmate, and purpose of visit. It includes a feature to capture a photo using a device camera, which is stored in the system for identification. Once completed, the data can be saved to support organized visitor management.

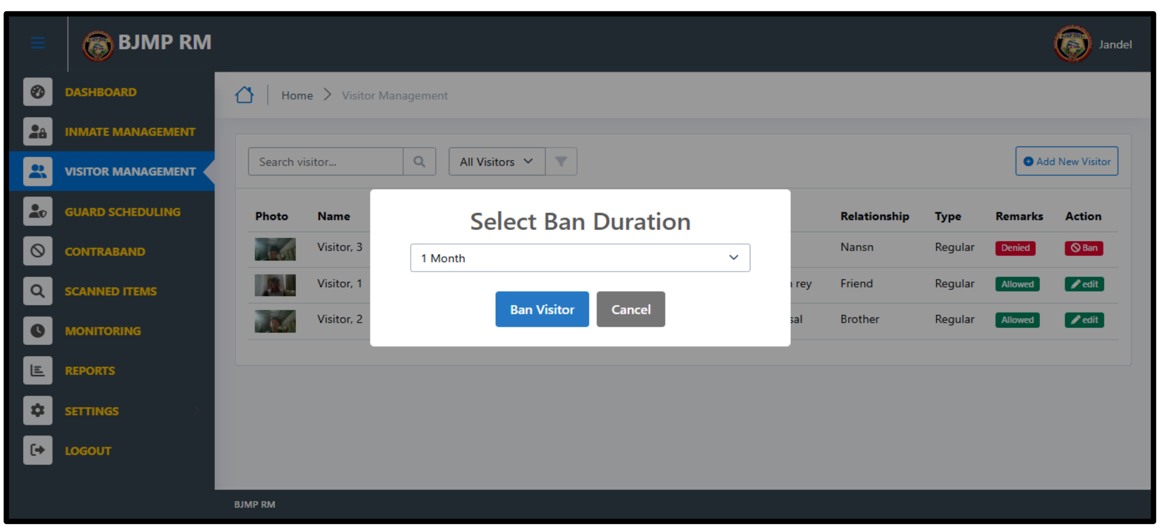

Figure 8. Ban Visitor

As shown in the preceding Figure 8, the interface displays the "Select Ban Duration" modal dialog that appears when an administrator initiates a visitor ban. The admin-only workflow allows selection of a specific ban period from the dropdown menu and then either confirming the restriction by clicking the blue "Ban Visitor" button or canceling the operation entirely.

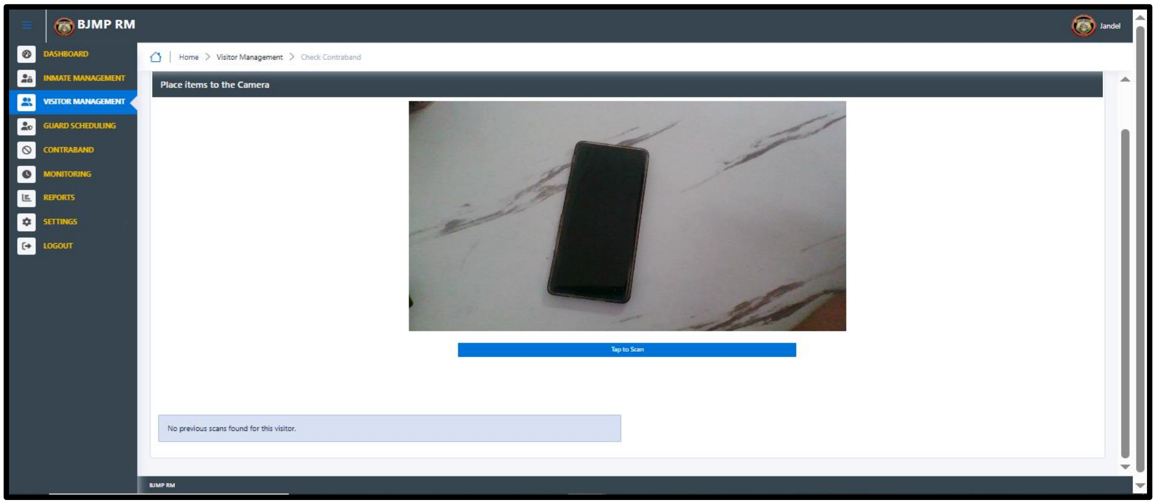

Figure 9. Contraband scanning,

The above Figure 9 shows the system activates the device camera after permission is granted, allowing it to capture and analyze items brought by the visitor. After analysis, the guard or staff of the day determines whether to accept or deny the visitor's entry based on the scanning results.

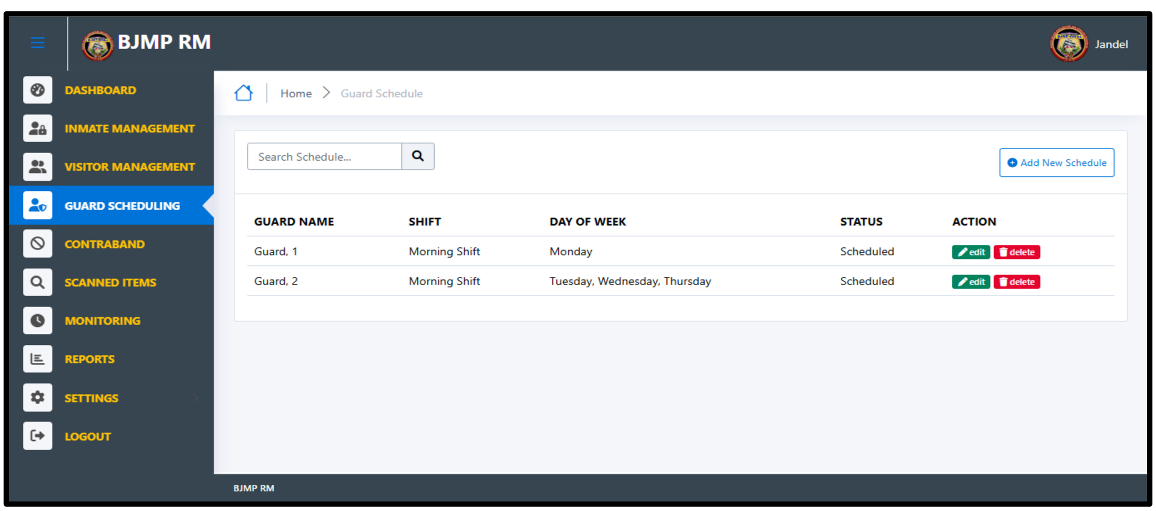

Figure 10. Guard Scheduling

The preceding Figure 10, shows a guard scheduling system. The interface displays a table with columns for the guard's name, shift, day of the week, status, and actions that can be taken, providing a clear overview for managing assignments and can able to login only if it has an assigned scheduled.

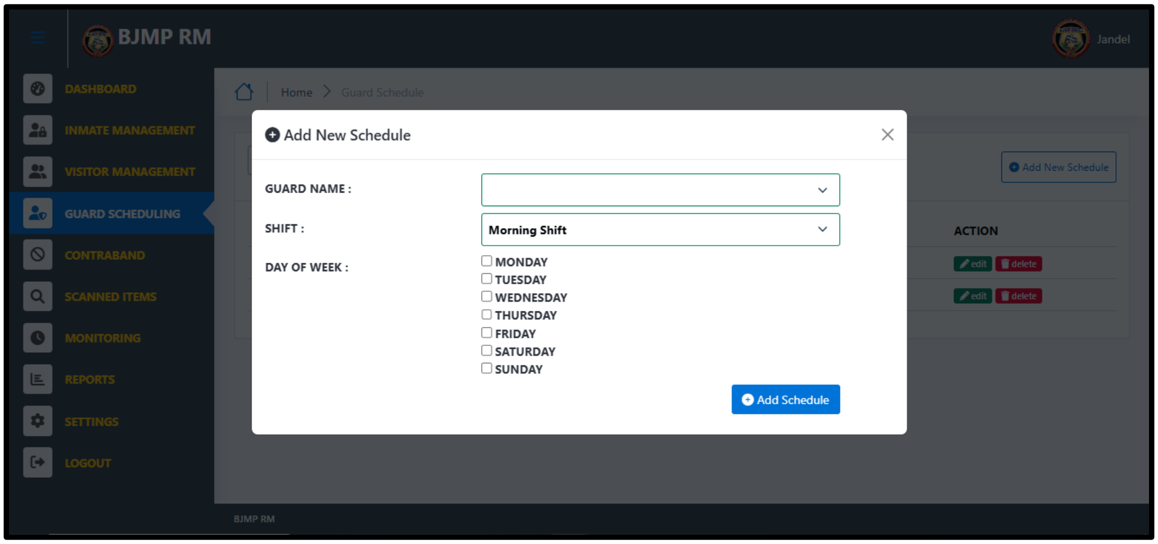

Figure 11. Add New Guard Schedule

The above Figure 11 displays information for adding a new schedule for a guard and assigning them to either a morning or night shift duty within a specific day of the week. The form ensures proper coverage and helps avoid scheduling conflicts by clearly outlining available shifts.

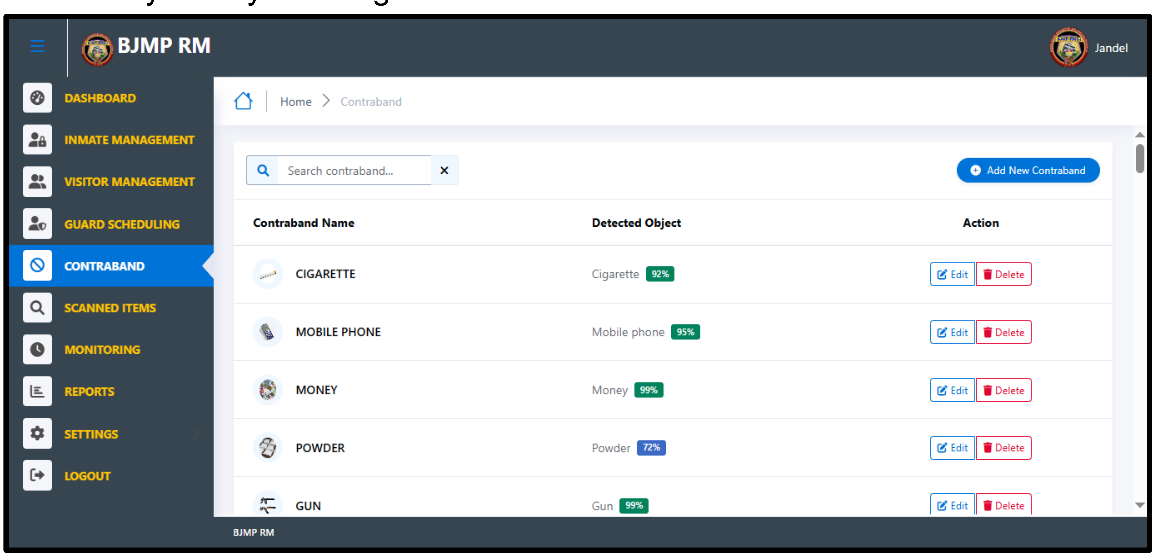

Figure 12. Contraband Management

The preceding Figure 12, displays a contraband management system. There are also buttons for adding new contraband items and viewing a list of contraband and allow admin to update or delete. The interface ensures that records of prohibited items are efficiently organized and easily accessible for monitoring and reporting purposes.

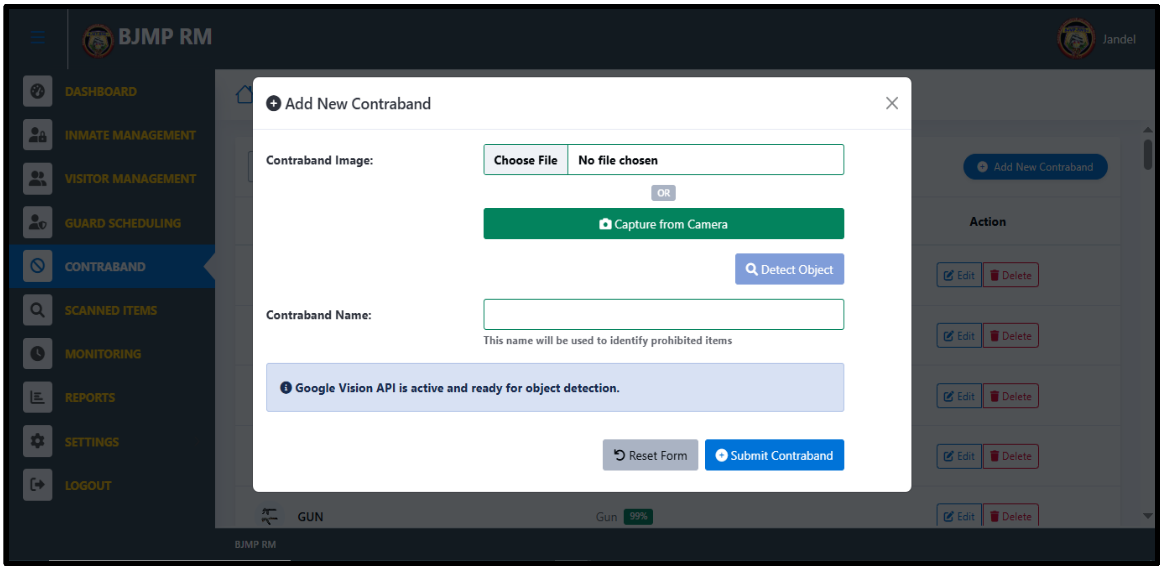

Figure 13. Add New Contraband

As shown in Figure 13 above, the interface displays the "Add New Contraband" modal dialog in the admin section. The form allows administrators to register prohibited items by either uploading an image file or capturing one directly using the Capture option. After obtaining an image, administrators can use the "Detect Object" button to leverage Google Vision API for automatic identification of the image especially the name.

The form includes a field for entering the contraband name, with a helpful note explaining that "This name will be used to identify prohibited items." The modal provides "Reset Form" and "Submit Contraband" buttons at the bottom for form management

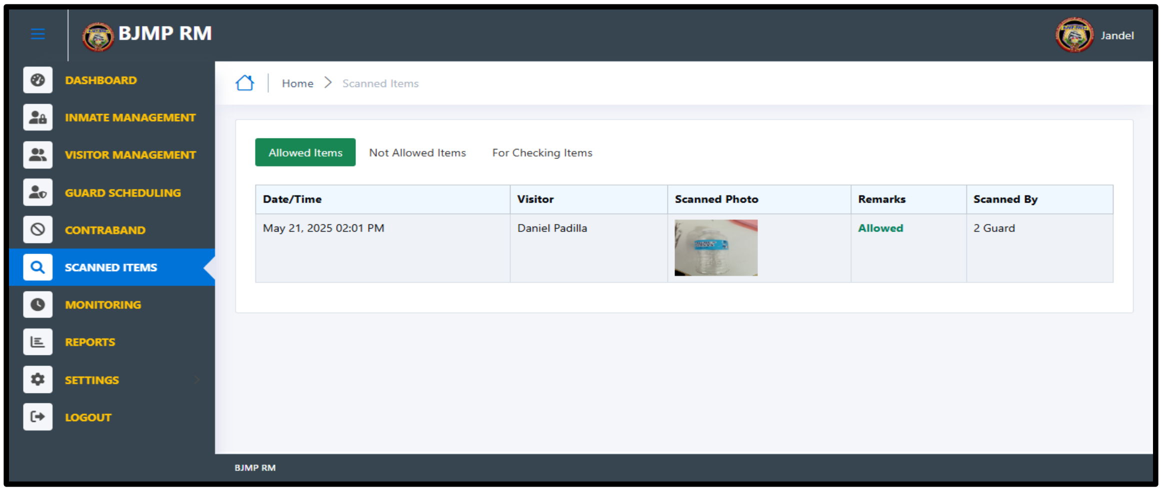

Figure 14. Scanned Items

In Figure 14 above, the Scanned Items interface displays a log of scanned visitor belongings with timestamps, visitor information, item photos, approval status, and scanning personnel. The interface includes filtering tabs for different item categories. This feature enhances monitoring efficiency by allowing quick access to item histories and decision outcomes.

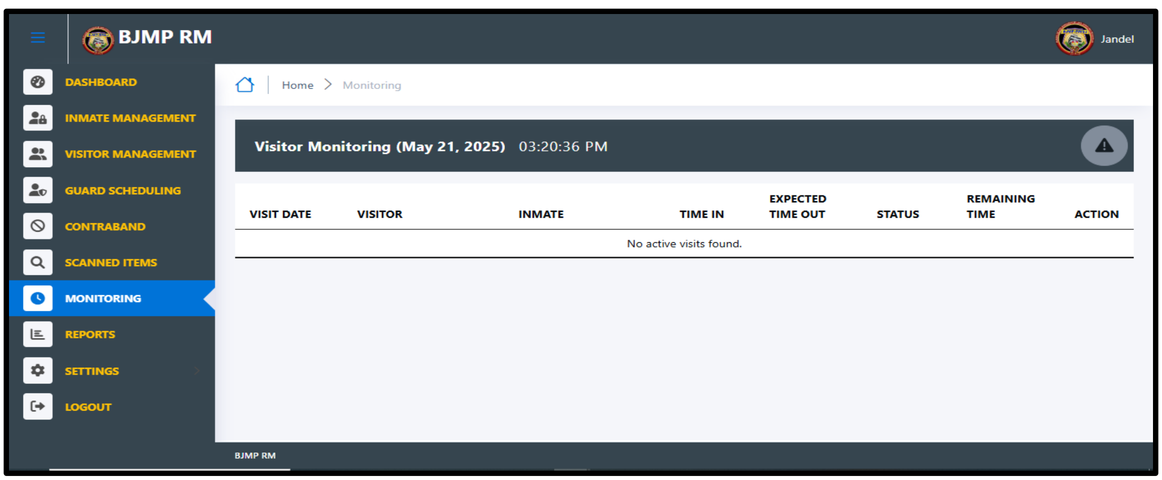

Figure 15. Visitor Monitoring

As shown in Figure 15 above, it displays the time each active visitor who enters the jail from type of visit and allows the admin to force check out the visitor if needed. This helps ensure accurate monitoring of visitor duration and supports security by managing overstays.

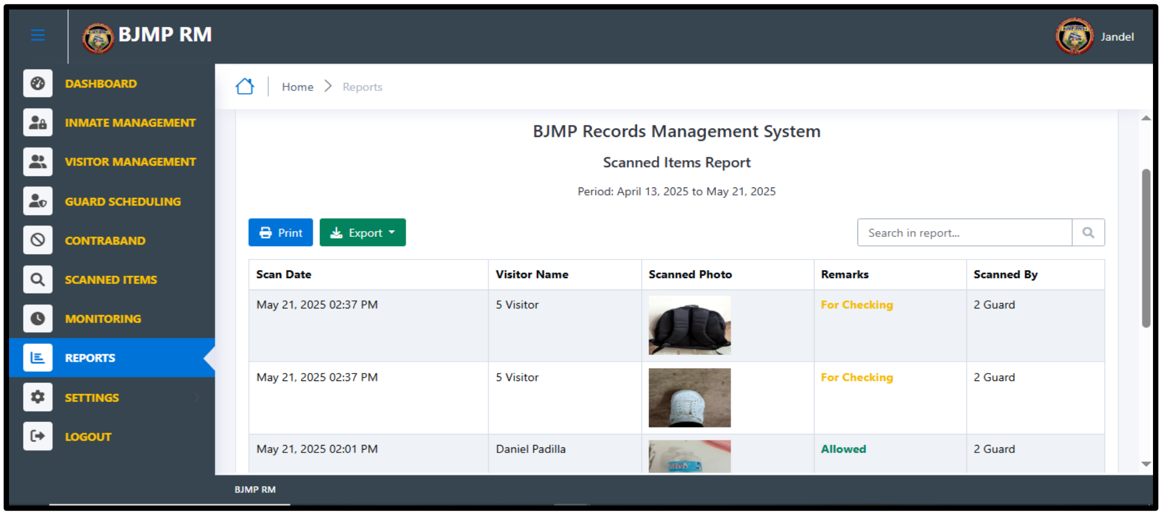

Figure 16. Reports

In Figure 16 above shows a reports section within a prison management system. The interface allows the admin to search for specific scanned contraband items, visitor's logs, and inmates for a specific time range, helping streamline data retrieval and auditing processes.

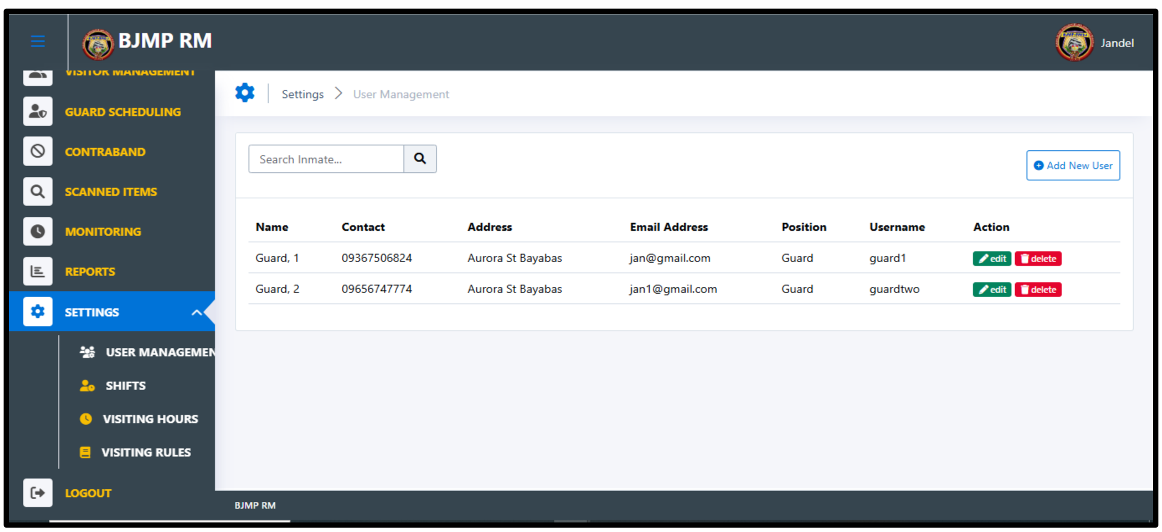

Figure 17. User Management

The above Figure 17 shows a user management section within a prison management system. It lists existing users along with essential details such as their full names, contact numbers, residential addresses, email addresses, designated positions, and system usernames. The interface allows administrators to edit or delete user accounts as needed, helping maintain accurate and up-to-date user records. Additionally, the system features a search bar for quickly locating specific users and a clearly labeled button for adding new user accounts. This section plays a critical role in ensuring that only authorized personnel have access to the system, reinforcing role-based access control and contributing to secure and efficient system administration.

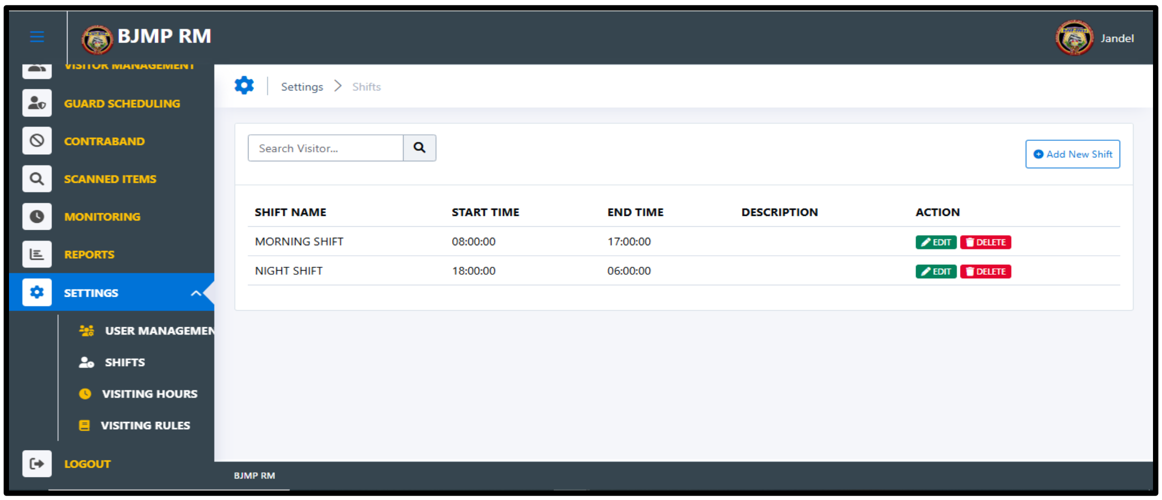

Figure 18. Shifts

In Figure 18 above, the shift management section is shown as part of the prison management system. This interface displays a detailed list of existing shifts, including each shift’s name, start and end times, and any additional descriptions provided. It features functionality for administrators to add new shifts, modify current ones, delete outdated entries, and search for specific shifts as needed. This tool is essential for organizing personnel schedules, helping ensure proper coverage across all hours of facility operations. By maintaining clear and structured shift data, the system supports improved staff coordination, reduces scheduling conflicts, and enhances the overall efficiency and security of the correctional facility.

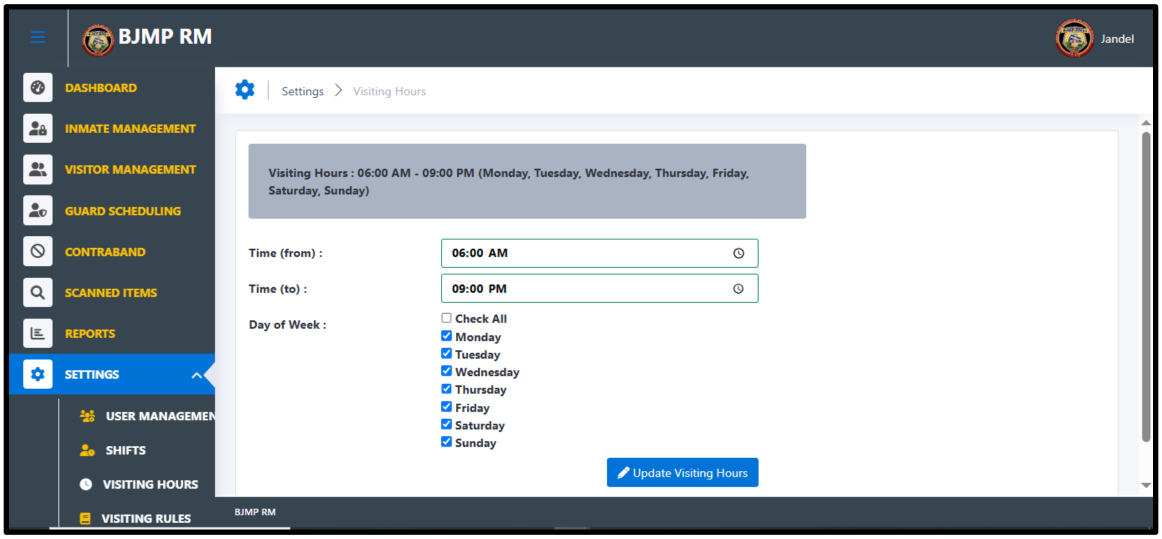

Figure 19. Visiting Hours

In this Figure 19 above, the interface allows the admin to set visiting times for specific days of the week and update them at any time. If no active visits are scheduled, the system displays a notification or blank state to indicate available slots. This feature streamlines scheduling, prevents overlaps, and supports secure visitor access based on institutional policies.

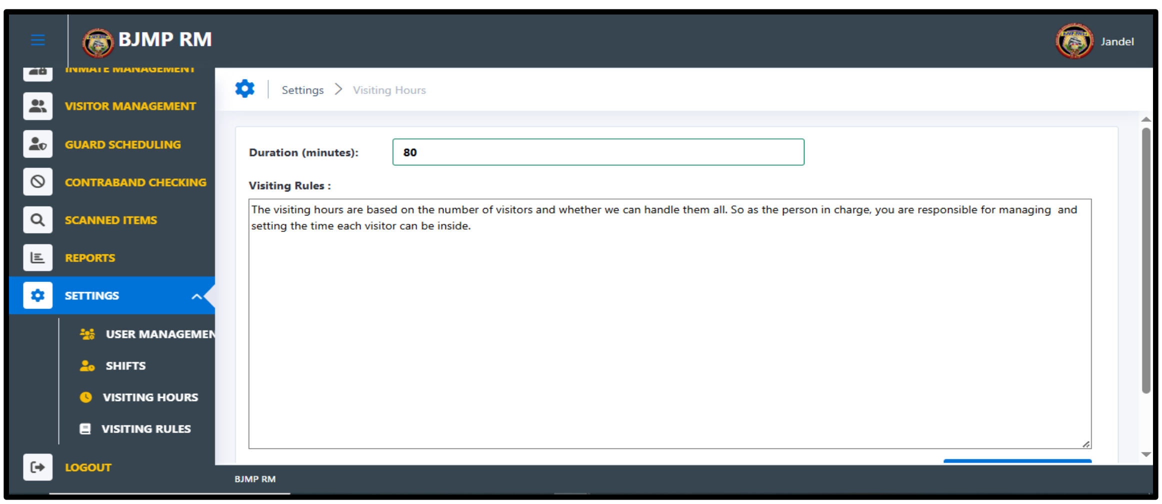

Figure 20. Visiting Rules

The above Figure 20 displays the interface for managing visiting rules within the prison system. This feature enables the admin to set specific time limits for each type of visit, such as legal, family, or emergency visits, ensuring that visitation remains structured and compliant with facility policies. The system also allows for easy updates or modifications to the rules, providing flexibility in managing visitation schedules based on operational needs or security concerns. Clear rule-setting enhances order, improves monitoring, and reinforces the institution's commitment to secure and fair visitation practices.

Staff

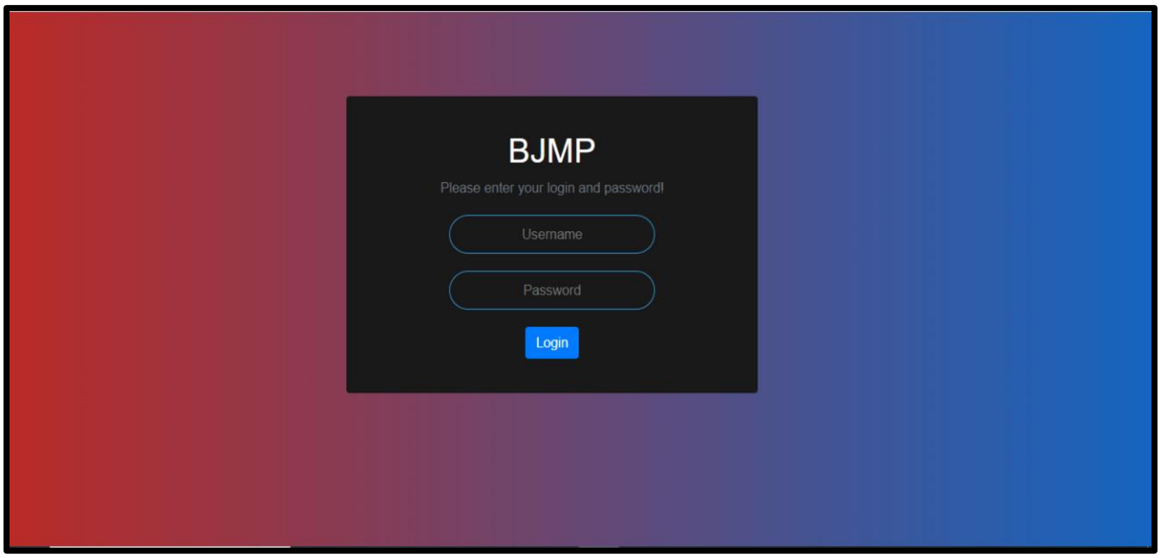

Figure 21. Log in interface of the Staff

As illustrated in the Figure 21 above shows the staff login interface features essential input fields, including a username and password section, along with a login button. This interface acts as the secure entry point for authorized personnel to access the system.

Staff must input their assigned credentials correctly to proceed. The login process helps ensure that only authenticated users such as jail officers or administrative staff ,can interact with sensitive data, such as visitor logs and contraband records. This mechanism also supports user tracking and audit trails, contributing to overall system security and accountability.

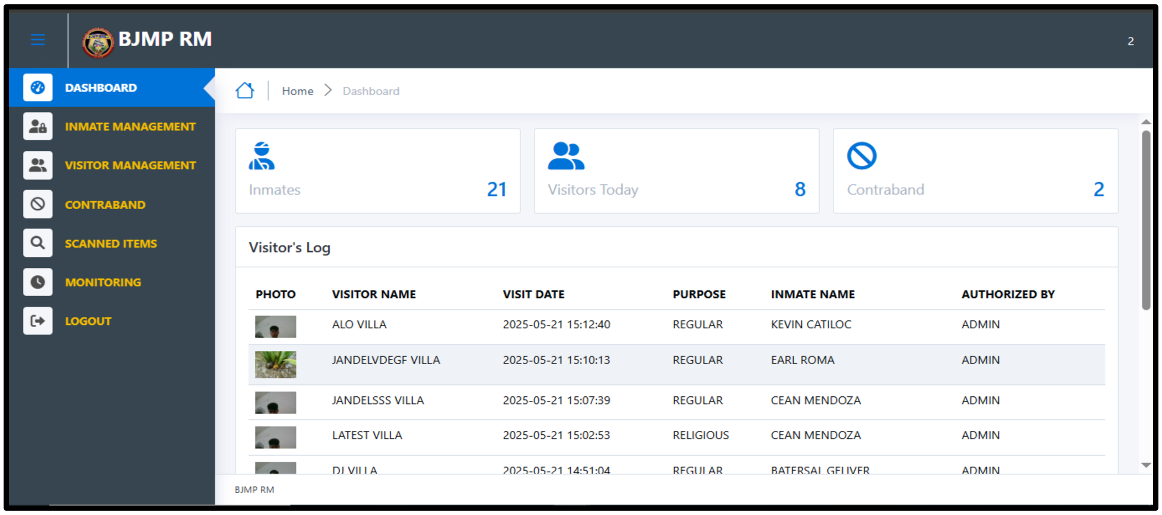

Figure 22. Dashboard of the Staff

In Figure 22 above, the interface displays the staff dashboard of the prison management system. It provides a real-time overview of key metrics, such as the number of registered inmates, visitors for the day, and detected contraband items. This centralized view helps staff monitor the facility's status and navigate easily to related modules for efficient task execution and improved awareness.

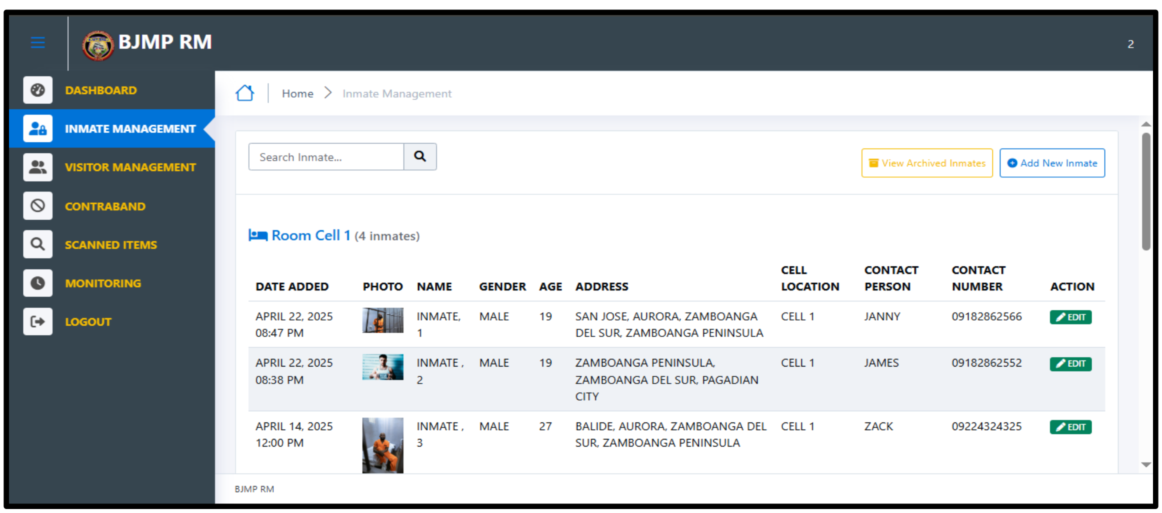

Figure 23. Inmate Management

The above Figure 23, shows the inmate management interface. The interface displays a list of inmates, with columns for their photo, name, gender, date of birth, address, cell location, contact person, and contact number. It also includes options to update or remove inmate records, helping ensure the database remains current and accurate.

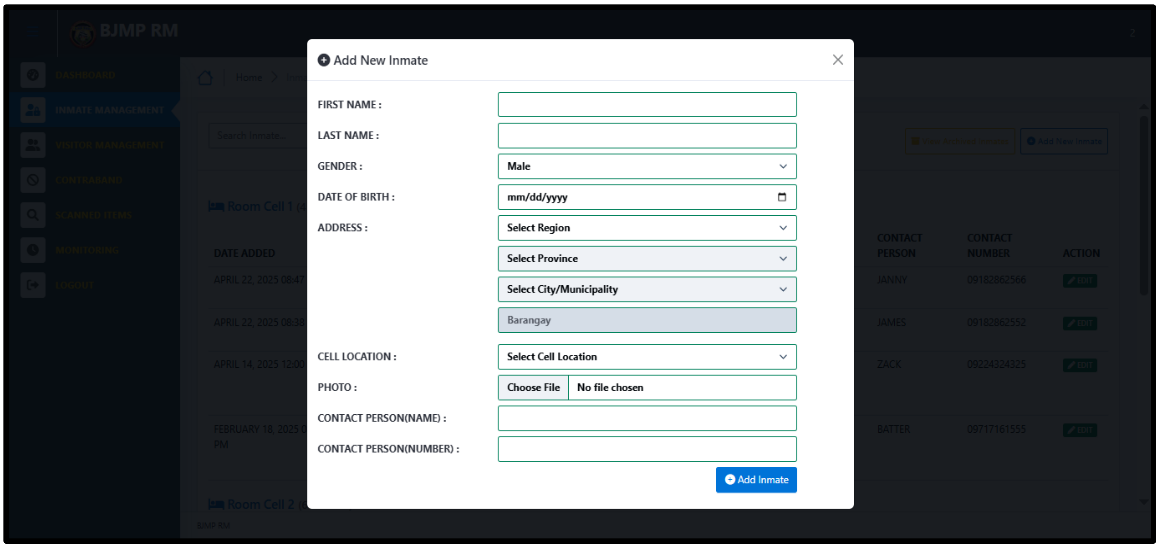

Figure 24. Add Inmate

In this Figure 24 above, the interface shows the process of adding a new inmate into the system. It includes fields for the inmate's personal details such as name, date of birth, gender, address, cell assignment, photo upload, and emergency contact.

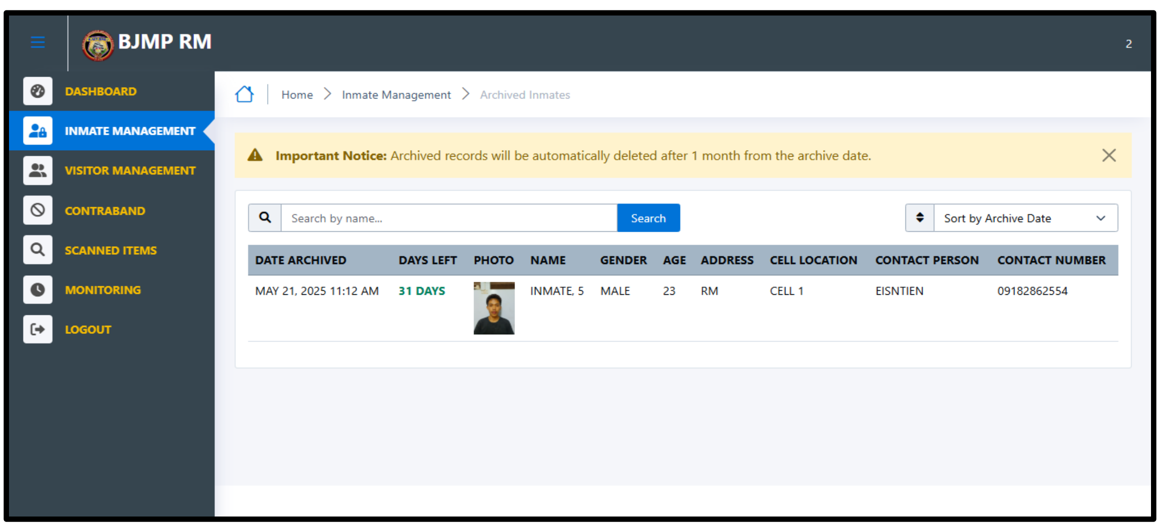

Figure 25. Archived Inmates

The above Figure 25 displays temporarily removed inmate records with archive dates, deletion countdown, and complete inmate information. A warning notice indicates records will be automatically deleted after one month, and staff can search and sort archived entries as needed.

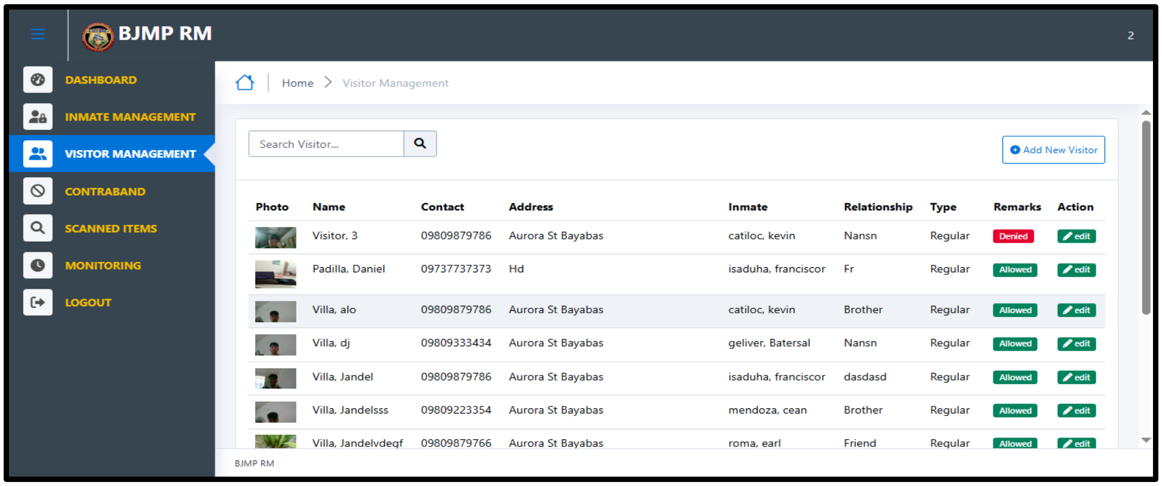

Figure 26. Visitor Management

In Figure 26 above, shows a visitor management system. The interface displays a list of visitors, with columns for photo, name, address, the inmate visiting, the relationship of visitor to the inmate, the type of their visit, and actions that can be taken (editing).

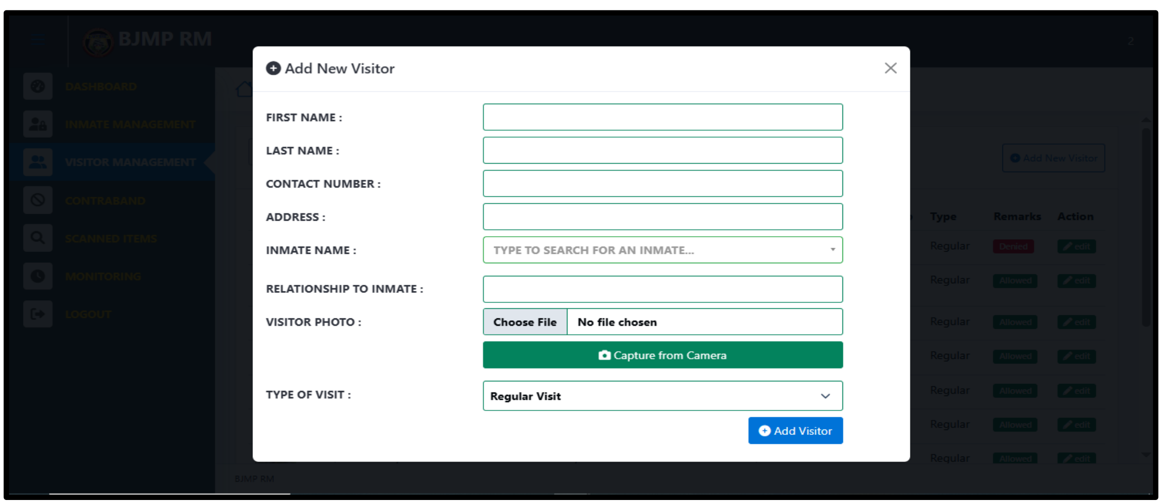

Figure 27. Add Visitor

The above Figure 27 displays a registration form where staff can input essential visitors' details such as the visitor's name, relationship to the inmate, and reason for visit. It also includes a feature to capture an actual photo using a device camera, which is saved for identification and security purposes. This helps ensure proper documentation and efficient tracking of all visitors entering the facility.

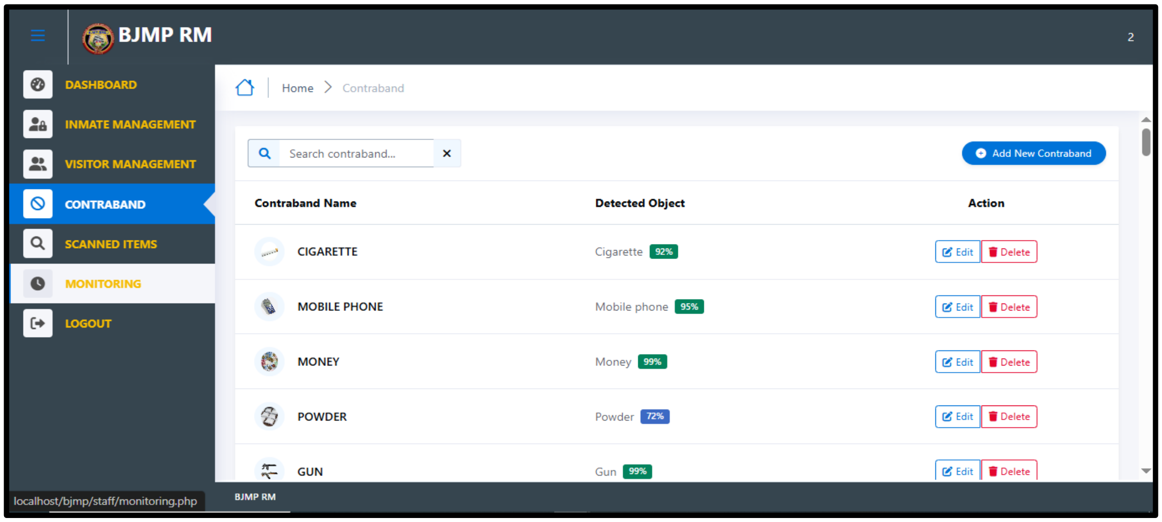

Figure 28. Contraband management

In Figure 28 above displays the contraband management interface, allowing staff to manage records of confiscated or prohibited items. It includes options to add new entries, view the existing list, and handle item details, supporting efficient tracking and reinforcing facility security.

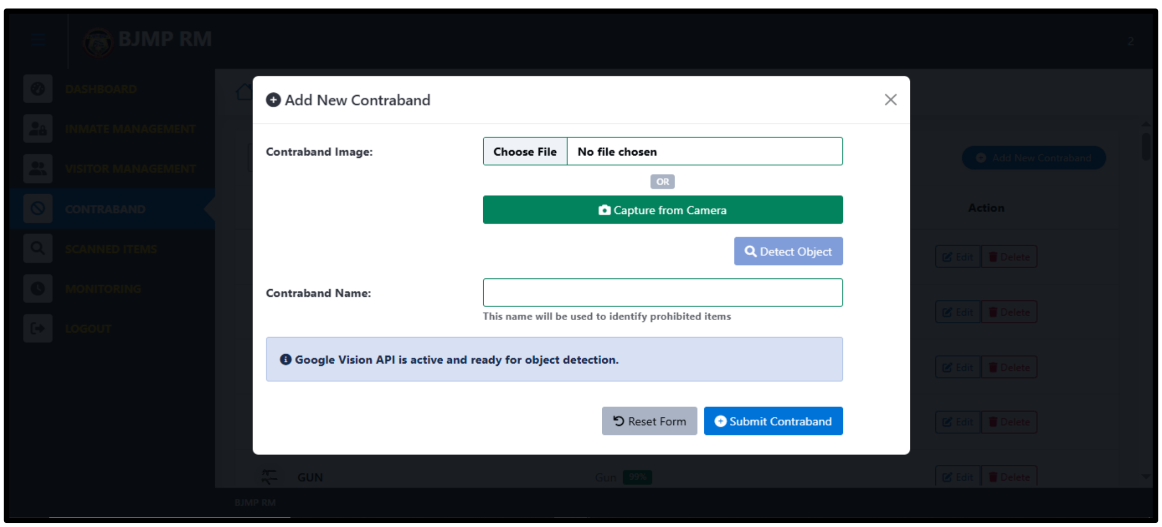

Figure 29. Adding New Contraband

In this Figure 29 above, the staff can upload or capture a contraband image, use the "Detect Object" feature to identify items in the photo, enter the contraband name, and submit the entry. The modal also shows the status of the Google Vision API and provides options to reset the form or submit the contraband.

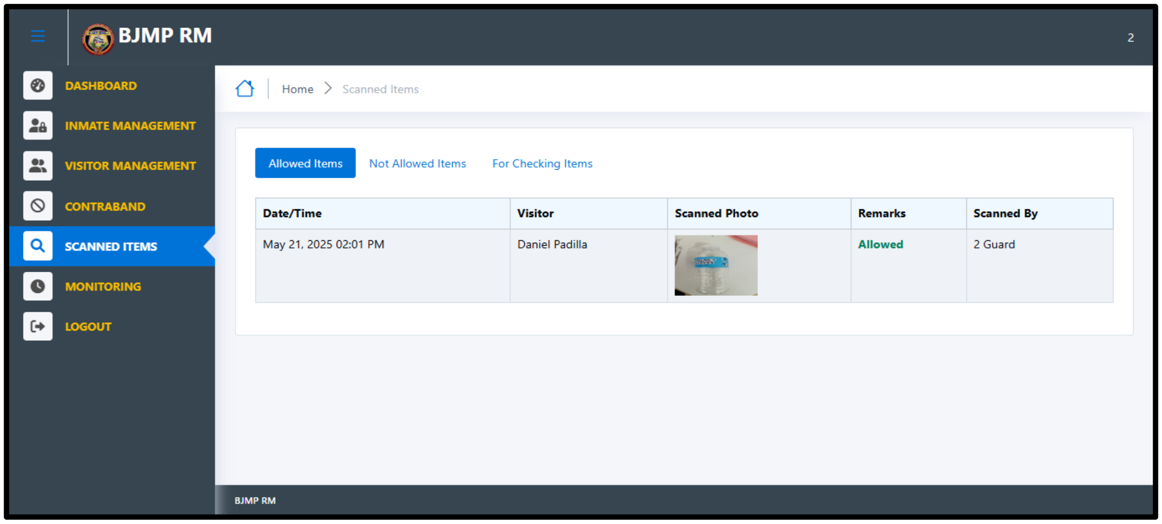

Figure 30. Scanned items

The above Figure 30, display and allow staff to view a list of scanned items, including the date and time of the scan, visitor name, scanned photo, remarks (such as "Allowed"), and the name of the guard who performed the scan. Tabs allow filtering between allowed, not allowed, and items for checking.

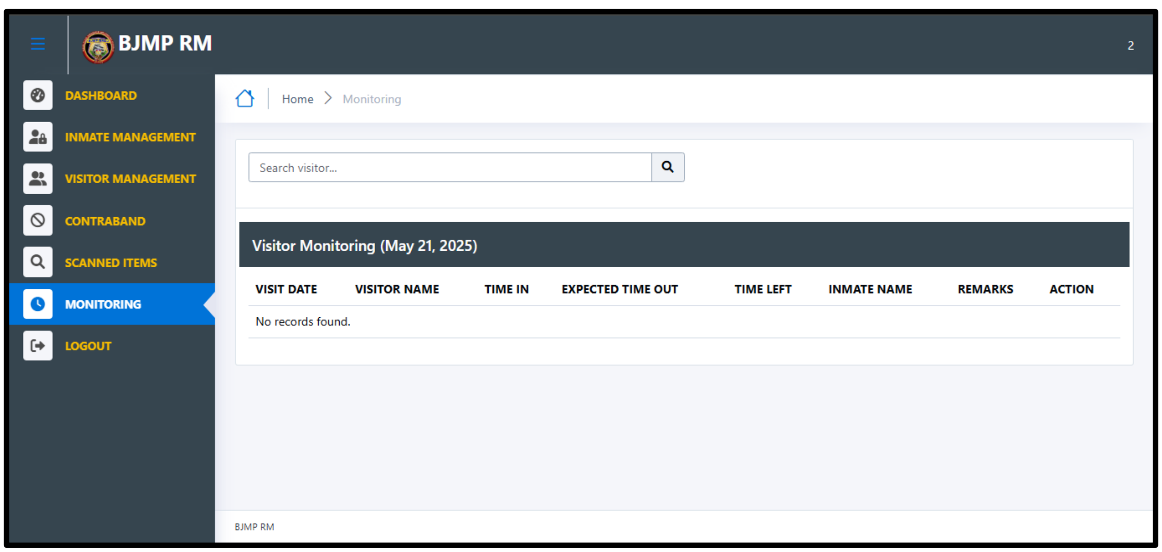

Figure 31. Visitor Monitoring

As shown in Figure 31 above, displays the remaining visit time for each visitor based on the assigned visit type, ensuring that the duration of each visit complies with facility regulations. The interface allows staff to monitor visit status in real time, including check-in and expected check-out times. It provides visual indicators or countdowns for ongoing visits, helping personnel manage time-sensitive access. Additionally, once a visit is completed or exceeds the allotted duration, staff can facilitate the "visit completed" process through the system. This feature improves overall visitor supervision, prevents overstays, and supports orderly operations within the facility.

CHAPTER V

SUMMARY OF FINDINGS, CONCLUSIONS AND RECOMMENDATION

This chapter provides an overview of the study's outcomes, including its summary, conclusions, and recommendations. The research is centered on the pursuit of a more precise method of monitoring visitors and detection in the correctional institution. It offers recommendations from the researchers intended to provide valuable insights for future scholarly endeavors in various research pursuits.

Summary of Findings

The findings from the development and testing phases highlighted several important results that directly addressed the study's objectives:

1. Streamlined Visitor Log Monitoring System. The implementation successfully transformed traditional paper-based logbooks into an automated digital system. This transition significantly improved efficiency by reducing human error and simplifying visitor tracking processes. The system meticulously captured essential data about each visitor's entry and exit times, creating comprehensive visitor records that enabled security personnel to accurately monitor visitor flow and identify potential security risks in real-time.
2. Effective Contraband Detection Implementation. The system effectively implemented image-processing-based contraband detection using mobile cameras to capture images of visitor belongings. This technology proved to be a

significant improvement over traditional manual inspection methods, as it could systematically identify prohibited items through advanced image analysis. The integration of image processing technology enhanced the accuracy and effectiveness of contraband detection, providing a more reliable security screening process that reduced the likelihood of prohibited items entering the facility.

3. Improved Visitor Management and Security Protocols. The system successfully integrated an automated approach that enhanced both service efficiency and security measures. The user-friendly interface made it easy for security personnel to operate, with mobile devices that were straightforward to use for capturing images and an administrative dashboard that allowed for quick access to reports and logs.

4. Effective Offline Functionality. The system's offline capability enabled it to work effectively without relying on continuous internet access, making it especially suited for correctional facilities where internet availability might be inconsistent.

5. Comprehensive Platform Development. Overall, the developed system fulfilled the general objective by creating a comprehensive platform that effectively utilized image processing technologies while operating as an offline website. This implementation enhanced security protocols and streamlined visitor management processes in correctional institutions, demonstrating a successful technological solution to the challenges identified in the study.

Conclusion

The research conducted on the Visitor's Log Monitoring System and Contraband Detection demonstrates that the system is highly effective and aligns seamlessly with the BJMP Manual of Operations. Utilizing image processing technology and automated visitor tracking, the evaluation provided a comprehensive analysis of the system's performance across essential operational requirements, including security enhancement, visitor management, and contraband detection. These capabilities indicate that the system not only meets but exceeds the operational requirements of correctional facilities, facilitating a significant transition from manual to automated processes, which enhances both efficiency and accuracy in security operations.

The successful implementation of this integrated system highlights its potential advantages in incorporating advanced technological solutions within correctional facility operations. Its design and flexibility serve as a valuable model for other correctional institutions aiming to leverage technology to improve their effectiveness and maintain rigorous security standards. The system's offline functionality further ensures its practicality in various environments, regardless of connectivity challenges.

Overall, this Visitor's Log Monitoring System and Contraband Detection stands as a testament to how effective technological integration can transform correctional facility operations, ensuring better resource management, enhanced security protocols, and operational excellence while maintaining alignment with established BJMP guidelines. The combination of automated visitor tracking and advanced contraband detection represents a major advancement in creating safer, more secure environments for both staff and inmates in correctional institutions.

Recommendations

Based on the study's findings, the following recommendations are offered to enhance future improvements and research. These aim to strengthen system functionality, user engagement, and overall monitoring effectiveness.

1. Enhance AI-driven contraband detection by developing a more extensive dataset with real-world contraband images to improve model accuracy and adaptability.
2. Explore alternative contraband detection technologies, such as thermal imaging, RFID tracking, or chemical sensors, to complement image-processing methods.
3. Implement a multi-factor authentication system for both visitors and staff, such as biometric scans, QR codes, or one-time passwords, to enhance security access control.
4. Investigate the use of blockchain technology for secure logging and tamper-proof recordkeeping of visitor data and contraband reports.
5. Develop a mobile application version of the system to allow correctional staff to monitor logs and receive alerts remotely for improved response time.
6. Implement an integrated visitor scheduling system that allows for pre-booking of visits, helping to manage facility capacity and reduce waiting times while maintaining security protocols.
7. Establish a comprehensive data analytics module to identify patterns in visitor behavior and contraband attempts, enabling proactive security measures and resource allocation.

8. Develop an automated notification system for inmates about upcoming visits to improve facility operations and inmate preparation for scheduled visitations.

Given the emphasis on security and efficiency in the findings, future research should focus on conducting regular security vulnerability assessments to identify potential risks within the Visitor's Log Monitoring System and Contraband Detection solution. By implementing these recommendations, correctional facilities can build upon the existing strengths of the system, leading to improved security protocols, better resource management, and enhanced overall performance in maintaining a safe environment for both staff and inmates while adhering to BJMP operational standards.

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APPENDICES

APPENDIX A. Letter to The Office

ZAMBOANGA DEL SUR PROVINCIAL GOVERNMENT COLLEGE
AURORA MAIN CAMPUS
COLLEGE OF COMPUTING STUDIES

May 13, 2024
BJMP Regional Director
Baranggay Lenienza, Pagadian City

Dear Sir/Ma'am,

I am writing to formally request approval to conduct research in the BJMP facilities under your jurisdiction. The research aims to gather information regarding ways of inspection of the prisoner's visitors, allowed food, scheduling of visit, time duration of visitors, and other related security policy from any staff or officer in BJMP institution.

The undersigned researcher, Jandel A. Villa and Dulce Marie Maglasang along with Kimberly Jean Pabutawan (Researcher's Adviser) and Delmer Mondido. (Dean of the College), seek permission to conduct the research specifically in Ramon Magsaysay District Jail. The details of the research request are as follows:

Researcher's Information:

- Name: Jandel A. Villa
- Contact Number: 09367506824
- Email Address: jandelamarovilla@gmail.com

Research Title: Visitors Log Monitoring System and Contraband Detection in Correctional Institution Specific Jail Facility: Staff Office
Number of Clients Needed: 1-2 staffs
Tentative Schedule for Submission of Approved Research Output to BJMPRO-IX:
May 27, 2024

We assure you that the research will be conducted with utmost professionalism and in adherence to all BJMP guidelines and regulations. We are committed to ensuring the confidentiality and integrity of the research process.

We kindly request your prompt consideration and approval of this research request. Your support and cooperation in this matter are highly appreciated. Thank you for your attention to this request. Should you require any further information or clarification, please do not hesitate to contact me at the provided contact details.

Respectfully yours,

JANDEL VILLA

DULCE MARIE MAGLASANG

Recommending Approval

KIMBERLY JEAN B. PABUTAWAN
Research Adviser

DELMER MONDIDO
Campus Dean

CURRICULUM VITAE

JANDEL VILLA

Student

PROFILE

BSIS student with a strong sense of responsibility and organization. Eager to gain my first work experience and apply my knowledge in a professional setting.

CONTACT ME

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Purok Rubia,
Bayabas Aurora
Zamboanga del Sur

EDUCATION

ZAMBOANGA DEL SUR PROVINCIAL GOVERNMENT COLLEGE
Information Systems career, in progress.

ZDSPGC
2020-2025

LANGUAGE

Cebuano
Tagalog
English

COMPUTER SKILLS

MS word and PowerPoint
Excel
Adobe Photoshop

INTERESTS

Dance
Gaming

MOTTO

Learn, Adapt, and Keep Moving Forward.

DULCE MARIE MAGLASANG

Student

PROFILE

Bachelor of Science in Information Systems student with a responsible and detail-oriented mindset. Excited to gain hands-on experience and apply my skills in a professional environment.

CONTACT ME

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Purok 6, Eastern Bobongan Ramon Magsaysay Zamboanga Del Sur

EDUCATION

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LANGUAGE

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English

COMPUTER SKILLS

MS word and PowerPoint
Adobe Photoshop

INTERESTS

Dance
Playing Musical Intruments

MOTTO

Let go of who you think you're supposed to be and embrace who you are.